



Lighting the way forward: The bright future of photonic integrated circuits

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ABSTRACT

Integrated optics, a key photonics technology, has major implications for telecommunications, sensing, and computing. By integrating optical elements like lasers, modulators, and detectors onto a single chip, it enhances performance while reducing the size, weight, and power of optical systems. With potential applications in 5G, data centres, quantum communication, and AI, integrated optics is crucial for future communication technologies. Its synergy with quantum information and machine learning opens innovation pathways. This review explores recent advancements in optical platforms, highlighting ongoing research and developments that are driving technological progress in this field. The analysis provides an in-depth scrutiny of recent developments, shedding light on the dynamic terrain of optical innovations and highlighting the strides made in pushing the boundaries of technology.

1. Introduction

Photonic integrated circuits (PICs) offer a paradigm shift within the domain of circuitry, presenting advantages that position them as a superior alternative to traditional electronic circuits [1]. Unlike electronic circuits that rely on the movement of electrons through conductive materials, PICs leverage photons or light particles to carry and process information. This fundamental difference imparts several key benefits. Firstly, PICs enable data transmission at the speed of light, significantly outpacing electron-based systems [2]. This inherent speed advantage results in higher data transfer rates and reduced latency, crucial in applications such as high-speed communication networks and data-intensive computing [3–5]. Additionally, PICs exhibit lower energy usage, as photons encounter less resistance/scattering compared to electrons, leading to more efficient energy use. Furthermore, PICs are less susceptible to electromagnetic (EM) interference and can transmit signals over longer distances, than their electronic counterparts, without significant signal degradation [6]. These characteristics make PICs particularly well-suited for operations where speed, energy efficiency, and reliable signal processing in long-distance communication are paramount, marking notable progress over electronic circuits in these specific fields [7,8].

Photonic circuits stand at the forefront of driving optical computing into the future, leveraging the distinct characteristics of light to revolutionize information execution and transmission [9]. Within the field of optical computing, photonic circuits intricately employ essential

components for instance waveguides (WGs), modulators, and detectors to deftly manipulate light signals [10]. WGs adeptly guide and channel light, and modulators finely control the intensity or phase of the light signal, while detectors skilfully convert light signals back into electrical signals for subsequent execution [11]. The notable capability of photons to traverse at the speed of light and concurrently transmit information across multiple wavelengths is instrumental in crafting highly parallel and efficient optical computing systems. Consequently, the potential of photonic circuits lies in the transformative redesign of computing architectures, offering promising solutions to overcome traditional electronic circuit limitations, and ushering in an era of more potent and energy-efficient computing solutions [5,12].

PICs usher in a new era of chip-based photonic functionality, enabling the seamless integration of optical components onto a single chip. Notably, PICs demonstrate significant versatility, allowing integration not only within the field of photonics but also harmonizing with microelectronics through both hybrid and monolithic approaches [13–15]. The ongoing trend towards elevated levels of integration favours the widespread embrace of silicon (Si) photonics, particularly in utilizations such as LiDAR. The integration of PICs with other technologies, especially microelectronics, presents a paradigm shift that transcends the conventional boundaries of individual components. This convergence holds the promise of enhanced functionality, reduced form factors, and increased efficiency. In the context of LiDAR deployments, this integrated approach yields a substantial price- and volume-scale effect, paving the way for cost-effective mass production. The result is

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an accelerated integration of LiDAR technology into the mobility space, aligning with the industry's drive towards achieving widespread adoption of autonomous and connected vehicles [16].

The future of PIC technology holds great promise as it continues to revolutionize various industries [17]. PICs are poised to be transformative in multiple fields, from enabling lightning-fast communication networks to enhancing computing capabilities, sensing technologies, and healthcare operations [18]. For instance, in communication, PICs could facilitate the expansion of high-speed optical communication systems (OCSs), boosting data transfer rates significantly. In computing, the integration of photonics into processors could lead to unprecedented processing speeds and energy efficiency [9,10]. In healthcare, PICs might revolutionize medical imaging by enabling highly sensitive and real-time diagnostic tools [19–21]. As researchers and engineers delve into advanced materials and fabrication methods, the miniaturization of photonic components is expected to result in more compact and efficient devices, with utilization ranging from portable consumer electronics to space exploration equipment. Moreover, the integration of photonics with evolving technologies for instance quantum computing and artificial intelligence (AI) opens new frontiers, allowing for the development of sophisticated and powerful systems [22]. With ongoing innovations, we anticipate a future where PIC technology becomes ubiquitous, fostering innovations that profoundly transform the way we communicate, compute, and interact with the world around us.

Integrated optics, despite its immense potential, has faced many challenges in achieving widespread availability for several fields [23, 24]. One significant obstacle is the complexity and precision required in the manufacturing processes of PICs [25–27]. Fabricating these circuits demands sophisticated procedures and materials, often leading to higher production costs [28]. Additionally, the incorporation of various photonic elements on a single chip requires a high level of expertise in multiple disciplines, from material science to optical design. Standardization and scaling of production have also been burdening with barriers to overcome, as the technology involves intricate designs that are not easily mass-produced. Furthermore, the education and training required for specialists to work effectively with integrated optics have been a limiting factor in the broader adoption of this technology. As R&D efforts continue to address these challenges, and as demand grows, it is anticipated that integrated optics will become more accessible and find its way into diverse applications, ultimately contributing to the proliferation of this transformative technology [18].

This paper unfolds in a planned sequence, commencing with Section 2, where an insightful exploration of the global PIC market is presented. This section precisely delves into the dynamic landscape of growth and investments within the photonic industry, offering a comprehensive understanding of the prevailing market trends. Moving forward, Section 3 delves into the forefront of technological development, unveiling the major optical platforms currently under R&D. This section provides a subtle viewpoint on the cutting-edge innovations that are shaping the future of photonic integration. Transitioning to Section 4, the narrative seamlessly shifts to the potential applications of PICs. This part delves into the diverse realms where PICs exhibit significant promise, offering readers a sight into the multifaceted dimensions of their practical applications. Section 5 takes centre stage by addressing challenges and their corresponding solutions related to the development of future PICs. This critical examination provides a balanced view of the hurdles faced in the integration of PICs and, in turn, explores innovative solutions that pave the way for advancements in the field. To end, the final Section 6 contains insightful concluding remarks. This section encapsulates the essence of the paper, summarizing key findings, highlighting overarching themes, and potentially suggesting avenues for future research.

2. Global PIC market

The PIC Market is poised for remarkable growth, projected to reach

an estimated value of US\$ 4.72 billion by 2029, unveiling a compelling Compound Annual Growth Rate (CAGR) of 17.7 % throughout the forecast period as shown in Fig. 1 [29]. This robust expansion underscores the increasing significance and widespread adoption of PICs across diverse technological landscapes. As these circuits continue to redefine the boundaries of information and communication technology, the market's upward trajectory reflects their pivotal role in driving innovation and efficiency across various industries.

The global PIC market is witnessing robust growth, driven by the proliferation of information and communication technology, the consolidation of telecommunications and networking equipment, and the growing need for high-bandwidth (BW) transmission devices [30, 31]. A foremost factor driving the market growth is the increasing adoption of PICs in optoelectronic (OE) devices attributable to their low power consumption, higher efficiency, and operating speed. The increasing adoption of fibre optic technology is also driving the demand for PICs globally [29].

However, challenges such as the lack of processing and fabrication standards pose a hindrance to market expansion. Additionally, the absence of digitalization in some developing and underdeveloped countries limits market growth to some extent. Nevertheless, the surge in data volumes and the corresponding demand for data centres present well-paid opportunities for major manufacturers in the global market. The indium phosphide (InP) segment dominates the market in terms of raw materials, exhibiting substantial revenue share. Geographically, North America holds a significant share, driven by advanced IT and telecommunication infrastructure, while the Asia Pacific market is projected to witness the highest growth, fuelled by the existence of major electronics component producers and fast developments in telecom set-up [29].

3. Material platforms for integrated optics

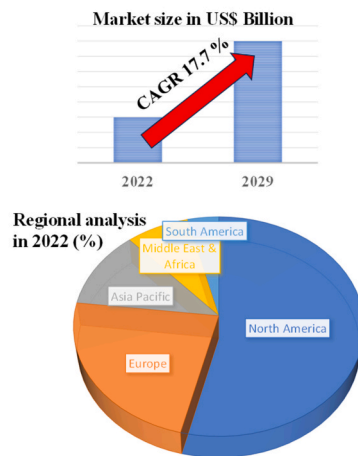
In the dynamic landscape of IO, a multitude of optical platforms are positioned to play a decisive role in determining the trajectory of this rapidly evolving technology. In this section, we delved into a comprehensive exploration of both extensively researched and cutting-edge platforms within the field of integrated optics. We covered well-established technologies as well as emerging platforms, providing a nuanced understanding of the diverse landscape within this field. By examining the latest expansions alongside the tried-and-true solutions, we aim to offer a holistic perspective that captures the dynamic evolution of integrated optics platforms.

3.1. Silicon platform

Si photonics stands out among other optical platforms due to its unique set of characteristics that make it a compelling choice for a variety of implementations. One key advantage is its compatibility with existing Si-based electronic technology, allowing for seamless integration with microelectronics on a single chip. This integration facilitates cost-effectiveness and scalability, making Si photonics an economically viable option. Moreover, Si's high refractive index enables the miniaturization of optical components, leading to compact and efficient devices [32]. Si photonics also exhibits low signal loss and supports a broad wavelength range, contributing to enhanced performance in communication systems. The ability to leverage standard semiconductor fabrication processes further streamlines production and reduces manufacturing complexities. Additionally, Si photonics offers the potential for active elements, such as modulators and detectors, which can be integrated directly into the Si platform [33,34]. These characteristics collectively position Si photonics as a versatile and powerful technology, distinguishing it as a frontrunner within the field of optical platforms.

Si photonics harbours the immense potential to revolutionize diverse technological sectors, heralding a new era characterized by developments in high-speed communication, computing, and sensing

Global PIC Market



Main performers

-Infina Corporation	-Kaia Corp
-NeoPhotonics Corporation	-POET Technologies Inc
-Huawei Technologies Co., Ltd	-II-VI Incorporated
-Finisar Corporation	-Intel Corporation
-Broadcom Inc	-Agilent Technologies, Inc
-Luxtera, Inc	-Aifotec AG
-OneChip Photonics	-Alcatel-Lucent SA
-Intel Corporation	-CyOptics Inc
	-Enablence Technologies Inc

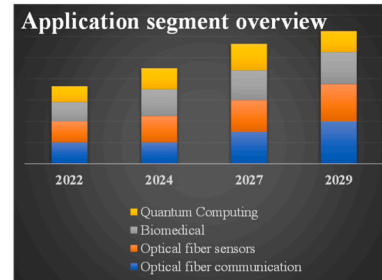


Fig. 1. PIC global market. Inspired by Ref. [29].

[35–37]. Through the integration of optical components on Si substrates, Si photonics is positioned to assume a crucial role not only in data centres and telecommunications but also in broader functionalities. In data centres, this technology facilitates the creation of energy-efficient and high-BW interconnects, streamlining data transfer processes and minimizing latency. Within the telecommunications realm, Si photonics can elevate the performance of optical communication networks, permitting augmented data rates and enhanced capacity. Beyond these domains, Si photonics emerges as a contender in cutting-edge fields like quantum computing, offering efficient solutions for quantum communication and interconnects [3]. Furthermore, its potential extends to biosensing and medical diagnostics, leveraging its high sensitivity to detect and analyze biological substances [38]. The ongoing advancements in R&D in Si photonics promise transformative changes as its integration permeates various sectors, paving the way for an interconnected and technologically advanced future [15].

Significant progressions have recently marked the advancement of the Si photonics platform, particularly revealing remarkable devices and systems for telecommunications [18]. This momentum has spurred heightened research interest in the platform's applicability beyond telecommunication, especially in areas necessitating compact devices and high production volumes. Sensing emerges as a key domain where Si photonics demonstrates promise, playing a crucial role in biological analysis [26,33,35] (e.g., molecule screening, biomarker investigation, reactions monitoring, and drug expansion) besides environmental surveillance [39] (e.g., pollutant and toxin tracking, food safety). However, the traditional integrated Si photonics devices, notably the standard silicon-on-insulator (SOI) strip WGs based on a 0.5 μm wide and 0.22 μm high Si core, may not always offer the optimized solution for specific functionalities. To address this, employing sub-wavelength grating (SWG) WGs becomes a highly promising approach, utilizing a segmented WG core to customize optical properties effectively [40,41]. In contrast to traditional continuous high refractive index cores, SWGs comprise periodic small Si blocks interleaved with a lower refractive index material (i.e., the upper WG cladding, as depicted in Fig. 2 (a) [42]. Governed by the Bragg condition, the periodicity of these blocks allows for tailored light propagation without sacrificing low-loss characteristics, offering a versatile solution for application-specific needs. In Ref. [42], the design and empirical assessment of SWG-based rings implemented on SOI chips were introduced without upper cladding, enabling their utilization as sensors as shown in Fig. 2 (b). The rings produced through fabrication exhibit Q-factors spanning from approximately 1000 to 6,000, contingent upon the chosen SWG parameters. Through a precise comparison of empirical and simulated data, it was

anticipated that the sensitivities surpass 383 nm/refractive index unit (RIU) in water and 270 nm/RIU in air. These findings underscored the remarkable potential of the fabricated rings for sensing purposes, emphasizing their suitability for diverse environments, and indicating promising avenues for practical implementation (see Fig. 3).

3.2. Polymer platform

Polymer-based active and passive optical modules have demonstrated significant potential in facilitating economically viable optical systems, particularly crucial for cost-effective uses such as Fibre-To-The-Home (FTTH) networks [49–51]. The adoption of a low-temperature process and a wet-coating method across large areas, compatible with various substrates, emphasizes the key prerequisites for achieving mass production feasibility [28,52,53]. Notably, polymer WG materials exhibit high transparency, ensuring absorption losses are minimized to be lower than 0.1 dB/cm at all critical communication wavelengths [54]. This transparency, coupled with the refractive index of most polymers (~ 1.5 – ~ 1.7), closely matching that of glass optical fibres (~ 1.5 – ~ 1.6), facilitates minimal Fresnel reflection losses at interfaces, particularly advantageous in the Butt-coupling arrangement of WGs and I/O fibres. Beyond passive modules, the exploitation of Thermo-Optic (TO) and Electro-Optic (EO) effects in polymers has enabled the development of active devices [55]. EO polymer-based optical modulators present several benefits over standard lithium niobate (LN) modulators, owing to the distinctive properties inherent in polymer materials [56]. This combination of attributes positions polymer-based optical modules as promising candidates for advancing the efficiency and affordability of optical systems in wide-ranging functionalities.

A cutting-edge innovation in polymer optical interconnect materials has been achieved with the development of Refractive Index Contrast (RIC) polymers, specifically engineered for a diverse array of photonic interconnect applications [50]. These RIC polymers boast a remarkable tunability of refractive index within the range of $n = 1.42$ to 1.56. The precision in tuning the refractive index contrast (Δn) is achieved through a combination of composition adjustments and controlled ultraviolet (UV) exposure. What sets RIC polymer interconnects apart is their capability to directly pattern WGs in dry films, eliminating the need for any wet or dry etching processes. This innovation offers versatility by enabling access to various photonic platforms, including Si photonic chips, ion-exchange glass optical substrates, and arrays of optical fibres. A groundbreaking achievement is demonstrated through the implementation of a maskless lithography approach, marking the first realization of efficient single-mode polymer interconnect

fabrication. This innovative practice exhibited low-loss adiabatic coupling, showcasing approximately 1.5 dB loss at 1.55 μm , particularly noteworthy is its seamless integration with ion-exchange WGs through the formation of grayscale tapers. This breakthrough not only expanded the horizons of polymer-based photonic interconnects but also set a new standard for fabrication methods within the domain of OCSs.

Polymer-based optical circuits offer a versatile platform applicable to communication, sensing, and optical signal processing, capitalizing on the diverse optical properties of polymers, both linear and nonlinear. Addressing the historical challenge of effectively coupling optical fibres with polymer WGs, Kumar et al. presented a novel wafer-scale vertical coupling scheme, illustrated in Fig. 2 (c) [43]. This innovative approach integrated a silicon nitride (Si_3N_4) grating coupler with an inverse taper, effectively connecting an optical fibre to an SU-8 polymer WG. The effectiveness of this method was evident in achieving a notable maximum coupling efficiency (CE) of -3.55 dB in the C-band and -2.92 dB in the L-band, with respective 3-dB BWs of ~ 74 nm and ~ 80 nm. The study encompassed a thorough exploration, covering detailed design and simulation, precise assembly processes, and comprehensive characterization outcomes. By introducing a scalable and efficient surface grating method, this scheme emerged as a promising pathway for advancing polymer PICs with improved performance and practical usability.

3.3. GaN-on-Si platform

The emerging GaN-on-Si platform represents a groundbreaking avenue for achieving photonic integration, capitalizing on the exceptional photoelectric and electrical performance inherent in GaN-based materials [44,57]. Drawing parallels with the InP-based approach, the utilization of a GaN-on-Si platform in photonics integration offers a comprehensive array of active-passive optical devices [17,58]. Leveraging the abundance of large-sized Si wafers and the favourable thermal conductivity of the substrate, this platform holds the promise of delivering cost-effective and high-power-density solutions. Additionally, the GaN-on-Si platform enables the fabrication of high-performance electronic devices that can seamlessly integrate with the PIC, resulting in an advanced OE integrated circuit [59].

It is noteworthy that PICs based on the GaN-on-Si platform can extend the light wavelength coverage from UV to NIR, thereby addressing the demands of visible light applications. This achievement is unattainable with InP-based photonics integration and monolithic Si photonics [60]. Despite the inherent Stokes shift, the GaN-based multiple quantum wells (MQWs) exhibit overlapping emission and absorption spectra in the action region [61]. Leveraging this overlap, the GaN-on-Si platform enables the realization of both light sources and photodiodes (PDs) using the same MQW diode structure. By incorporating GaN passive modules with MQW diodes on a single circuit, a plethora of PICs with diverse functionalities has been reported [62, 63]—these are distinguished by their simplicity, compactness, and flexibility as compared to other III–V integration systems.

A notable aspect of the GaN-based MQW structure is its dual functionality, which enables it to serve as an electro-absorption modulator. Such characteristics as low power dissipation, compact structure and relatively low driving voltage are evident when utilizing this technology [64–66]. Building upon this insight, a PIC was proposed as a comprehensive on-chip system, encompassing a light source, modulator, PD, WG, and Y-branch splitter. It was of paramount importance to note that all active elements within this proposed system adopted the same MQW diode structure. This implementation of a multifunctional MQW eliminated compatibility issues among discrete active devices and significantly moderated manufacturing complexity. A multichannel on-chip system was proposed which demonstrated indirect light modulation capability, a characteristic validated through various perspectives [17].

Successful demonstration of second-order optical nonlinearity in a Si architecture was achieved through the heterogeneous integration of

single-crystalline gallium nitride (GaN) on Si (100) substrates. By carefully tailoring GaN μ -rings for dual resonance around 1560 nm and 780 nm, efficient and reconfigurable second harmonic generation at 780 nm was realized [44]. The wideband transparency window of GaN, spanning UV, visible, and IR wavelengths, positions this platform as a promising avenue for on-chip optical wavelength generation across the far IR to near-UV range, facilitated by second-order nonlinear processes mediated by the χ^2 parameter. A vigorous bonding process was employed to create photonic assemblies in gallium nitride (GaN) thin films atop SiO_2 on Si substrates (GaNOI) [44]. This process utilized crystalline GaN with a 1.8 μm thickness deposited on Si (111) wafers from Nitronex Corporation, as illustrated in Fig. 2(d). To facilitate bonding, a 1.5 μm layer of SiO_2 was deposited onto the GaN films via PECVD. The final structure comprised three layers, bonded to Si wafers exhibiting a (100) crystalline orientation and coated with a thermally grown oxide layer that is 1.8 μm thick. This bonding method resulted in the creation of an extremely robust substratum with minimal discrepancies and residual strain. The subsequent step was to eliminate the Si (111) bottom layer through SF_6 deep reactive ion etching. To achieve a polished surface, the newly freed GaN surface underwent chemical-mechanical polishing, with the thickness of the GaN layer being reduced to 400 nm [44].

3.4. Plasmonic platform

Optical WGs serve as the foundational element in both optical fibre and PIC technology [67–69]. While their passive guiding mechanisms are crucial, incorporating active photonic manipulation can significantly enhance their capabilities. As the demand for faster nanoscale WG circuits intensifies, characterized by stronger light-matter interactions and reduced power consumption, emerging applications necessitate innovative approaches. Nonlinear optics, with its femtosecond response times and terahertz BW, holds charm, albeit often requiring high powers or expansive footprints when relying solely on dielectrics. Plasmonic nanostructures offer a compelling solution, leveraging metals for truly nanoscale, energy-efficient nonlinear optics [69–72]. Initial enthusiasm has matured into cautious optimism, marking recent years with notable progress in leveraging nonlinear plasmonic WGs to enhance various photonic circuit functions across diverse application domains [73].

Plasmonic integrated devices represent a transformative paradigm within the domain of photonics, leveraging the unique properties of surface plasmon polaritons (SPPs) for advanced functionalities [74–76]. These devices exploit the interaction between EM waves and free electrons at metal-dielectric interfaces, allowing for nanoscale confinement of light and manipulation of its properties [77,78]. Plasmonic WGs, resonators, and modulators are integral elements in these integrated systems, facilitating the miniaturization of optical circuits and enabling the development of high-performance devices for applications such as sensing, imaging, and communication [75,79–82]. The ability to confine light at subwavelength scales opens avenues for enhanced light-matter interactions and the creation of compact, energy-efficient devices [76]. While challenges such as losses and fabrication complexities exist, the promise of plasmonic integrated devices in revolutionizing optical technologies underscores ongoing research efforts aimed at harnessing their full potential for next-generation photonic systems [83].

It is noteworthy that the realm of plasmonic for optical computing or information processing remains relatively least explored, with the predominant focus on elucidating fundamental principles and demonstrating proof-of-principle devices [79,80]. While significant strides have been made in comprehending basic plasmonic principles, most endeavours have thus far been dedicated to conducting empirical validations. The recent surge in studies on surface plasmons (SPs) has been largely driven by the field of optics, resulting in applications that closely mirror those in optics or photonics. These include WGs, devices for polarization conversion, and optical filters [84]. However, it is crucial to recognize that plasmons epitomize the extreme frequency limit of the

classical skin effect, a well-established concept in RF engineering. As a result, there is a growing investigation into the potential of plasmons in electric circuits, antennas, and electrical filters, which is leading to a novel perspective that extends beyond traditional optics and holds the potential for groundbreaking applications in information processing and computing.

Optical phased arrays have showcased their prowess in diverse applications, spanning from high-speed on-chip communications to the realization of vertical surface-emitting lasers. Despite the improvements achieved in Si photonics-based designs, a notable gap exists in the exploration of plasmonic-based optical phased arrays, which remain uncharted territory. Zeng et al. introduced and empirically validated a groundbreaking concept – a passive plasmonic optical phased array [85]. The proposed array not only demonstrated beam steerability in the far-field region but also achieved remarkably high directivity. Significantly, the research demonstrated the efficacy of RF phased array theory in characterizing the coupling dynamics of delocalized SPs within optical phased arrays. This connection between phased arrays operating at different wavelengths paved the way for various applications, such as on-chip optical wireless nanolinks, optical interconnections, and the incorporation of plasmonic lasers. This marked the beginning of a transformative era with expanded opportunities for advanced photonic systems.

Plasmonic hybrids hold promise for enhancing the capabilities of PICs, providing benefits like polarization rotation at the wavelength scale, optical volumes at the nanoscale, and improved nonlinearities [86,87]. Despite these benefits, most PIC-compatible configurations utilize solitary plasmonic elements, and more sophisticated circuits often require intricate, from-the-ground designs. In Ref. [45], a modular approach was introduced to the post-process commercially available SOI platform comprising a plasmonic rotator and a nanofocuser, both proficient in generating the second harmonic frequency of incoming light as shown in Fig. 2 (e). Parameters defining the geometry of the rotator-focuser are shown in Fig. 2 (f). In Fig. 2 (g), an SEM image reveals a fabricated Hybrid Plasmonic Integrated Circuit on an SOI WG, emphasizing the sharpness of the tip, approximately 10 nm, a dimension constrained solely by the gold (Au) grains formed during the evaporation process. Through empirical characterization of the SOI WG, the performance of each module was showcased, demonstrating intensity enhancements exceeding 200 in an estimated mode area of 100 nm² at a pump wavelength of 1320 nm. This modular strategy for plasmonic circuitry enhanced the practicality and broadened the scope of uses for this technology [45].

3.5. Hybrid integration

Hybrid optical platforms in integrated optics represent a cutting-edge approach that combines diverse materials and technologies to enhance the functionality and performance of photonic devices [88,89]. This innovative paradigm involves integrating different materials, such as Si, polymers, and III-V compounds, onto a single circuit to exploit their gains. By leveraging the unique properties of each material, hybrid optical platforms enable the creation of more versatile and efficient photonic circuits [90]. For instance, Si photonics provides excellent compatibility with existing semiconductor processes, while polymers offer flexibility and ease of fabrication [91]. The integration of these materials allows for the development of advanced photonic devices with improved performance, such as low-loss WGs, high-speed modulators, and efficient light sources [92]. Hybrid optical platforms are at the forefront of R&D in the field of integrated optics, paving the way for enhanced optical communication, sensing, and computing applications [93].

Expectedly, most materials utilized in photonic integration platforms, apart from silica, are also commonly employed in electronic integration. This is due to several factors. Firstly, a preference for photonic integration is observed to leverage commercially available

platforms that have already been recognized for electronic amalgamation. The commercial electronic market is currently surpassed by its photonic counterpart by nearly an order of magnitude. Therefore, there is a strong motivation to employ a universal platform for both photonic and electronic integrated circuits (EICs). This motivation arises from considerations within the manufacturing ecosystem. Additionally, electronic counterparts for functions like control, signalling, buffering and powering are often necessitated by photonic elements. In addition, semiconductors and related dielectric materials have emerged as highly promising options for both photonic and electronic elements. Examples include lasers, detectors, WGs, and transistors. The integration of semiconductor materials with dielectric materials has the potential for significant improvements in integration platforms, particularly when combined with other materials. Initially, the active exploration of III-V platforms, such as GaAs and InP compound semiconductors, was prompted by these considerations as attractive foundations for photonic and EICs in the 1980s [94].

The integration of photonics in a PIC platform extends beyond fundamental photonic waveguiding. It necessitates the ability to generate and amplify light [95], perform EO modulation [96], facilitate optical-electrical detection, engage in nonlinear optical frequency conversion, and process RF photonic signals. The III-V compound semiconductor platform, represented by materials like GaAs and InP, offered these functionalities as a PIC platform and excelled as an EIC platform. Although promising developments were observed in the research and development of optoelectronic integrated circuits (OEIC) in the late 1980s, the anticipated extensive market success of photonic-electronic very large-scale integration (VLSI) platforms did not materialise [97]. Interestingly, this failure was attributed to the same reason III-V transistors, including HEMTs, HBTs, and FETs, could not showcase competitive benefits against Si CMOS transistors in enabling large-scale, high-yield, and commercially viable EIC platforms. The hindrance was the absence of reliable and high-quality passivation methods for III-V materials.

Developing effective passivation materials for III-V CMOS devices has proven challenging in achieving the high quality provided by thermally oxidized SiO₂, known for its extremely low defects and high density. Si photonics has emerged as an alternative, utilizing Si as the WG core and thermally grown oxide as the under-cladding. This approach yields optical interfaces with remarkably low loss and defects, ensuring robust confinement of optical modes and assisting compact, low-loss bending of WGs. Moreover, the Si photonic platform shares similar fabrication processes with the Si CMOS platform, resulting in the commercial availability of Si CMOS photonic electronic-photonic-integrated circuits (EPIC). In comparison to the III-V semiconductor, the Si platform provides several benefits, including the widespread accessibility of 300 mm wafers, enhanced thermo-mechanical characteristics, plentiful material accessibility, and a fully developed end-to-end ecosystem covering design, testing, and packaging. Si photonics excels in large-scale integration, demonstrating superior integration density and yield, supported by effective passivation and low-loss optical confinement. The concept of a 'photonic Moore's law' applies to photonic integration, predicting advancements at an even faster pace than the electronic Moore's law [98].

Nonetheless, Si photonics lacked essential characteristics sought in photonics, including Pockel's effects, classical optical gain, magneto-optical effects, second-order optical nonlinearities, and non-volatile phase-change aptitudes within Si's optically transparent spectral range (1.2 μm –6.9 μm). A remedy to rectify these shortcomings involves heterogeneous integration, wherein elements made from distinct materials are integrated into a favored substratum, usually Si. Hybrid photonic integration, a variant of heterogeneous integration, attains more intimate and higher-density integration by creating optical WGs comprising heterogeneous materials, thus amalgamating diverse optical properties [99].

A fundamental framework for hybrid PICs should enable the

development of low-loss, compact, high-density, and high-yield PICs, while also facilitating cost-effective fabrication and packaging processes, and utilizing existing ecosystems [100]. In this context, Si-based photonic platforms, including Si_3N_4 and related WG PICs on Si, present themselves as promising candidates for wavelengths within the 1.2 μm –2.6 μm range (bounded by the transparent spectral ranges of Si and SiO_2). For shorter wavelengths, $\text{Si}_3\text{N}_4/\text{SiO}_2$ PICs on Si can broaden the spectrum to 0.4 μm –2.6 μm range. Alternative WG platforms, such as those based on AlN/GaN, have the potential to extend further into the ultraviolet (UV) range. While numerous III-V semiconductors can cover a wavelength range between 0.3 μm and 20 μm , they are associated with minimal yield and other suboptimal features when assessed to the $\text{Si}/\text{Si}_3\text{N}_4/\text{SiO}_2$ PIC platform on Si substrates.

Facilitating the efficient connection of III–V light sources with Si photonic circuits poses a pivotal challenge in integrated optics [101]. Crucial prerequisites encompass minimal coupling losses, a compact footprint, high assembly yield, and the capacity for automated processes in large-scale production. In Ref. [46], Billah et al. introduced photonic wire bonding as a solution, leveraging direct-write two-photon lithography for the in-situ assembly of three-dimensional freeform WGs between optical circuits. Illustrated in Fig. 2 (h) is the conceptualization of a hybrid photonic multi-chip module (MCM) facilitated by photonic wire bonds (PWBs), operating as a WDM transmitter. The SEM image of the laser chip interface is revealed in Fig. 2 (i). By conducting proof-of-concept experiments, the study established linkages between InP-based horizontal-cavity surface-emitting lasers and passive Si photonic circuits, achieving remarkably low insertion losses of 0.4 dB. The author asserted that this signifies the most effective interface between an InP light source and a Si photonic circuit demonstrated to date [46]. These experiments represented a noteworthy evolution in propelling photonic wire bonding toward becoming a versatile integration platform for hybrid photonic multi-chip assemblies. This facilitated the amalgamation of known-good dies composed of several materials into high-performance hybrid multi-chip modules [101].

3.6. Materials for nonlinear integrated optics

The majority of practical realizations of nonlinear integrated optics are based on lithium niobate (LN) material platforms. Thus, within our review, we have put the main emphasis on LN and its variants. However, to fully present the current landscape, we have dedicated a separate section to briefly summarize other technologies that also may provide strong nonlinear responses.

3.6.1. Lithium niobate

Integrated optics on the LN platform is a cutting-edge technology that leverages the exceptional properties of lithium niobate crystals to manipulate and control light at the microscopic level [102]. LN's excellent EO and nonlinear optical characteristics make it an ideal material for the amalgamation of various photonic elements on a single chip [103]. In integrated optics, various optical functions, such as modulators, detectors, and WGs, can be seamlessly integrated into a LN substratum [104]. This platform enables the miniaturization of complex photonic circuits, leading to enhanced performance, reduced power consumption, and improved reliability [105]. Moreover, the EO properties of LN allow for dynamic tuning of optical devices, making them adaptable to changing environmental conditions [11,104]. As a result, integrated optics on the LN platform holds great promise for applications in communication systems, sensing technologies, and quantum information processing [103,106]. The versatility and efficiency of this platform make it a key player in advancing the field of integrated optics for various real-world applications [106].

Within the field of the LN platform, Lithium Niobate-on-Insulator (LNOI) photonics stands out due to a significant index contrast compared to both air and the underlying silica, resembling the characteristics observed in SOI structures [107]. This distinctive feature has the

potential to facilitate the spatial confinement of light within micro/nano-scale photonic architectures. This confinement could prove beneficial for several applications, including high-density PICs and efficient nonlinear optical processes. It could also be advantageous for low-power-consumption surface acoustic waves and high-speed on-chip modulators. Over the past decade, there have been some promising developments in the nanofabrication of LNOI photonic structures. In 2007, Guarino et al. made a significant contribution to the field with their μ -ring resonator (MRR), which demonstrated a Q-factor of 4×10^3 [108]. Afterwards, in 2012, Hu et al. demonstrated an optical ridge WG with an optical loss of 6.3 dB/cm [109]. The evolution of LNOI technology has been extraordinary, with recent achievements including a microdisk resonator boasting a Q-factor of 4.7×10^7 [110]. Furthermore, the optical losses in single- and multimode LNOI WGs have been substantially minimized, reaching impressive values of 0.029 dB/cm [111] and 0.027 dB/cm [112,113], respectively. This progress underscores the growing potential and versatility of LNOI platforms in enabling advanced photonic functionalities with superior performance metrics.

In recent developments, thin-film LN has emerged as a highly promising platform for the implementation of PICs. A significant stride has been taken in the miniaturization of functional elements on this platform, marked by the introduction of high-speed LN EO modulators utilizing photonic crystal (PhC) nanobeam resonators as shown in Fig. 2 (j) [11]. These devices displayed notable altering efficiency, reaching up to 1.98 GHz/V, combined with a wide modulation BW of 17.5 GHz. Significantly, these modulators possessed a small EO modal volume of just 0.58 μm^3 . These devices enabled efficient EO control of high-quality photonic cavity modes in both adiabatic and non-adiabatic states, allowing for EO switching at 11 Gbps with exceptionally low bit-switching energy of 22 fJ. This demonstrated energy-efficient and high-speed EO modulation at the wavelength scale established a vital groundwork for the development of extensive LN PICs, with substantial implications for data communication, microwave photonics, and quantum photonics.

3.6.2. Other technologies for nonlinear integrated optics

Over the last few years, the LN has been very dynamically developed and recognized as a dominant platform for nonlinear integrated optics. However, several technologies are being developed for that purpose as well. Within this work, we classify the existing studies into two main categories: glass-based materials and semiconductors.

In general, glass materials, such as fluoride, tellurite, phosphate, borate or chalcogenide glasses, are well-established building blocks of any optical device and system, due to their isotropy, transparency and possible low cost of realization. Innately, glasses reveal also nonlinear optical response, that is relatively small in comparison to semiconductors but may be enhanced and adjusted by proper treatment, e.g., thermal poling [114], or introduction of nonlinear inclusions, e.g., nanocrystals [115]. On the other hand, glass composites based on heavy metal oxides (Sb, Bi, Pb, W, Ga, Ge, and Te) may also be designed to possess strong nonlinear response and low optical loss, which make them prospective candidates for optical switching applications [116]. Among others, GeSbS has been considered as one of the most promising chalcogenide glass materials for nonlinear applications, due to its large optical nonlinearity, i.e., $n_2 \approx 1.7 \times 10^{-18} \frac{\text{m}^2}{\text{W}}$, within mid-infrared telecommunication band [117]. As such, chalcogenide-based integrated optical components have been considered in a range of nonlinear applications, including optical switching, optical limiting and all-optical signal regeneration [118,119]. Doping glass matrix with metallic nanoparticle or semiconductor nanocrystals is yet another approach to enhance the nonlinear response of glass. The physical mechanism behind this lies in resonance phenomena occurring at optical frequencies near the electronic absorption edge [116]. As such, by doping with plasmonic inclusions or quantum dots, it is possible to achieve a

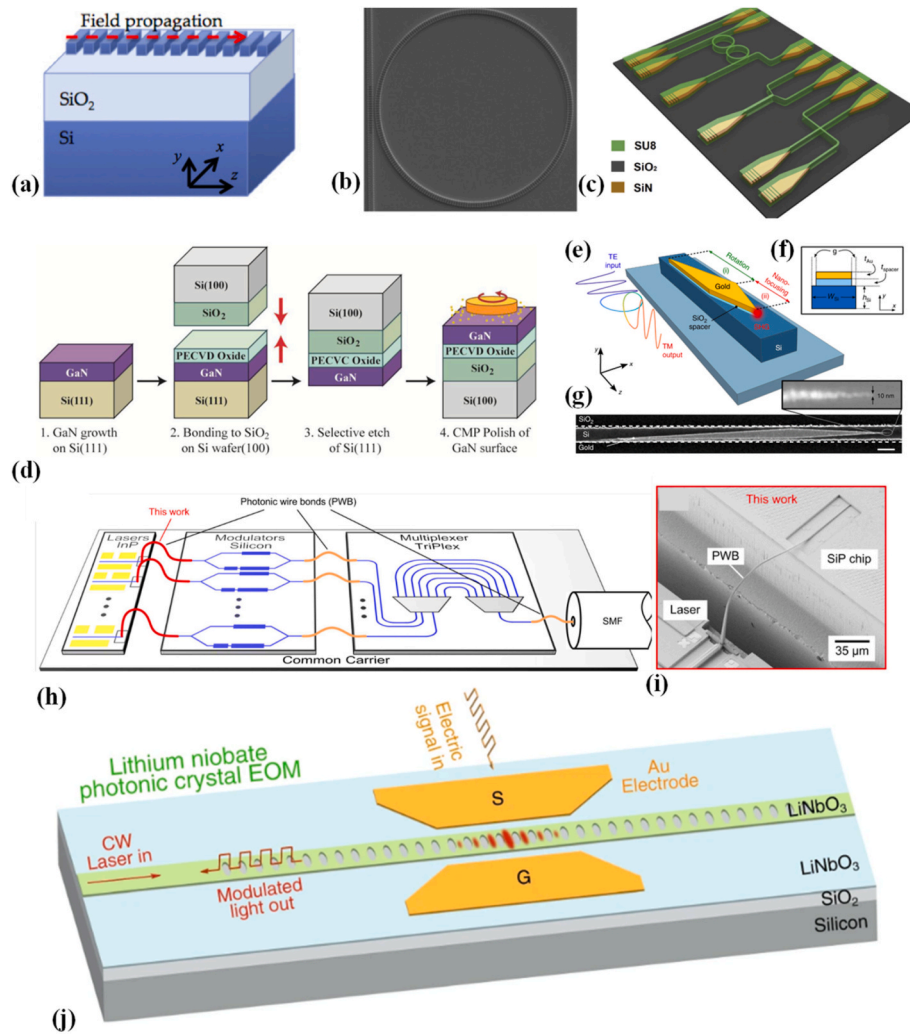


Fig. 2. (a) Schematic of SWG WG fabricated on SOI platform [42], (b) SWG RR structure based on SOI platform [42], (c) 3D conceptual design of a polymer photonic chip features the suggested configuration, incorporating SiN gratings, a spot size converter, and SU8 guides along with a ring structure [43], (d) GaN photonic circuits constructed on substrates known as GaN on SiO₂ on silicon (GaNOSI) [44], (e) configuration begins with an industry-standard TE ridge WG, succeeded by two consecutive plasmonic circuit modules: (I) an effective TE-photonic to TM-plasmonic rotator and (II) a nano-focusing tip [45], (f) parameters defining the geometry at the boundary of the rotator-focuser [45], (g) SEM image of the fabricated device [45], (h) a photonic multi-chip transmitter for WDM communications. The system harnesses Photonic Wire Bonds (PWBs) to amalgamate the distinct advantages of various photonic integration platforms: Direct-Bandgap (DFB) lasers are constructed on InP substrates, while Silicon Photonics (SiP) chips excel in implementing EO modulators. To achieve dense packing of optical channels, high-quality Arrayed WG Gratings (AWGs) are essential, and these are optimally realized on medium-index-contrast material systems like TriPlex [46], (i) SEM image displaying a laser-chip interface [46], (j) Schematic of the LN photonic-crystal EO modulator [11].

nonlinear response beyond that of bulk materials [120,121].

The other category is the semiconductor platform which is inextricably linked to the rise of integrated optics itself. From the very beginning of IO, semiconductors have been widely used in nonlinear integrated applications. As such, silicon, a flagship representative of semiconductors, offers a variety of possibilities, ranging from Raman amplification [122] to continuum generation [123] or four-wave mixing [124]. However, the use of silicon-based optics is still limited at telecom bands, due to the existence of two-photon and free-carrier absorptions [116,125].

The drawbacks of silicon in terms of nonlinear applications have been addressed by the heterogeneous integration of other material systems onto silicon substrates, such as silicon nitride (SiN), amorphous silicon and silicon carbide [116]. Even though SiN, regardless of stoichiometric and non-stoichiometric variants, reveals smaller Kerr nonlinearities than silicon, the wide transparency window makes it an interesting candidate for nonlinear signal processing applications. Until now, silicon-rich nitride and Si₃N₄ waveguides have been

demonstrated as wavelength converters, radio-frequency spectrum analyzers and optical parametric amplifiers [126,127]. On the other hand, the integration of hydrogenated amorphous silicon may provide, in comparison to its crystalline counterpart, stronger nonlinearity, which may be employed to obtain efficient four-wave mixing [128] or two-photon absorption [129]. Finally, among the possible material for integration, silicon carbide (SiC) has been employed due to its wideband transparency as well as strong second- and third- order nonlinear optical effects enabled by nanostructurization [116]. Until now, enhanced optical nonlinearity of silicon carbide has been successfully demonstrated in second harmonic generation [130] and applied in high-quality micro-ring resonators [131].

The other branch of semiconductor platform for nonlinear integrated optics is based on III-V materials. Among them, AlGaAs is the most intensively explored representative of this group. Due to a unique combination of strong Kerr nonlinearity, i.e., $n_2 \approx 20 \times 10^{-18} \frac{m^2}{W}$, nonlinear absorption, as well as low optical losses, i.e., $L_{min} = 0.4 \frac{dB}{cm}$ Available at telecom wavelengths [132], AlGaAs-based waveguide

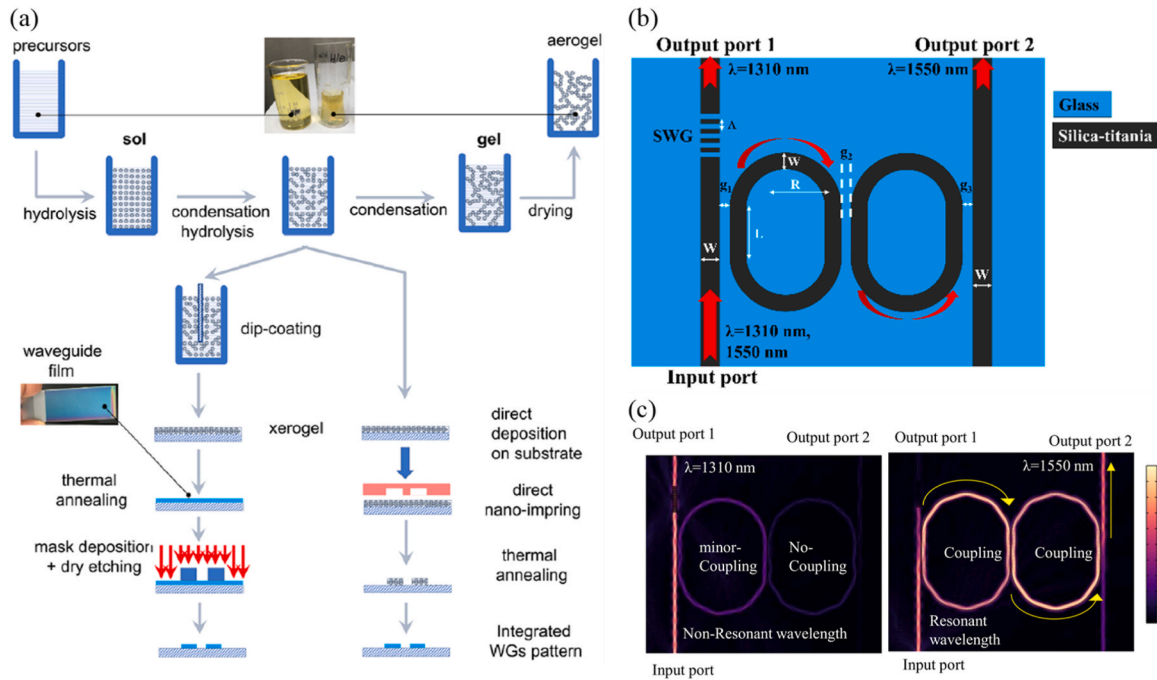


Fig. 3. (a) Schematic representation of modified fabrication process for $\text{SiO}_2:\text{TiO}_2$ integrated WG systems [47], (b) schematic of an integrated demultiplexer incorporating SWG and RR segments based on a modified silica-titania framework (c) Normalized E-field distribution in the demultiplexer at an operation wavelength of 1310 nm (left) and 1550 nm (right) [48].

systems offer immense application potential in all-optical signal processing. This platform has been considered in several structural varieties, e.g., strip-loaded waveguides, nanowire guiding layers, half-core waveguides, AlGaAs on insulators and suspended waveguides [133–135], which enabled several different practical realizations including, but not limited to wavelength converters, all-optical switching, time-domain multiplexers, nonlinear direction couplers, X-switches and polarization switches [132]. Latest achievements in that field cover also obtaining supercontinuum as well as frequency-comb in AlGaAs integrated systems [136,137]. Another promising III-V material that has been recently employed in integrated optics is gallium phosphide (GaP). Apart from large nonlinearity and transparency, the important feature of this material is the close lattice match with silicon, which may enable wafer-scale production. Up to date, there have been a few successful practical realizations of GaP devices working as frequency doublers and solid-state quantum emitters [138].

To sum up, the area of nonlinear integrated optics is being very dynamically developed field of immense potential, which will be even further growing in the near future due to its implications in quantum-related applications.

3.7. Low-cost alternative platforms

The strong business-oriented drive towards decreasing the overall cost of integrated optics fabrication stimulates the development and exploration of solutions, that are alternative to the mainstream technologies and may be more economically beneficial while maintaining high performance metrics and reliability [139].

To overcome the complexity and high production cost of the commercial integrated optics platforms, alternative approaches often involve inexpensive processes, such as sol-gel for material synthesis [140,141], to reduce overall fabrication costs. Among others, the silica-titania ($\text{SiO}_2:\text{TiO}_2$) platform has been considered as a low-cost alternative to the existing technologies [27]. Originally, silica-titania planar WGs operating within visible spectral range were demonstrated in the early 80s as a platform offering high mechanical durability and

insensitivity to chemical factors that may deteriorate their propagation properties [142]. The modified silica-titania framework, developed by Karasinski et al. involved the deposition of thin films via a simple and cost-effective combination of the sol-gel process and a dip-coating method [143], which are followed by nanoimprint lithography (NIL) to obtain the desired pattern. Despite the innate limitations of the framework, such as sensitivity to the aging process of sol-gel materials, it has been successfully demonstrated in practically realizable integrated systems composed of basic optical components operating in both VIS and NIR, such as photonic crystal WGs [47], RRs [144] and multiplexers [48].

Yet another promising low-cost technological platform is based on $\text{ZrO}_2\text{-SiO}_2$ organic-inorganic sol-gels [145]. The basic process for the proposed platform can be described with four basic steps: material synthesis, layer deposition, circuit printing, and thermal stabilization and typically involve two or more different precursors to obtain a mineral and an organic network [140]. Due to the choice of appropriate precursors, such as vinyl tri-alkoxysilane, these hybrid WGs may be fabricated at low cost and reveal relatively low scattering and intrinsic absorption within the second and third telecommunication windows, i. e., 1.6 and 6.9 dB/cm at 1310 and 1550 nm, respectively [146]. Until now, integrated optical systems based on the $\text{ZrO}_2\text{-SiO}_2$ platform have been successfully demonstrated in several applications including pH and gas sensors [147], water decontamination [148] and biosensors for health examining [145].

To sum up, the recent advancement in the field of inorganic-organic integrated optics enabled new applications and improvements which may increase its competitiveness to mainstream technologies. Additionally, due to the high potential of inorganic-organic platforms to reduce the environmental drawbacks for deployment, manufacturing and recycling issues of the current commercial integrated optics technologies, it has been perceived as a potential candidate for the next generation of sustainable integrated systems [149].

4. Reforming industries: the diverse applications of integrated photonic devices

Integrated optics stands as a pivotal technology, finding widespread application in a myriad of captivating fields, with ongoing exploration for future innovations as shown in Fig. 4. This section examines the indispensable roles played by integrated optics in a spectrum of cutting-edge applications. From the realms of autonomous vehicles (AVs) and augmented reality (AR) to the critical domains of data centres and communication, as well as the transformative landscapes of 5G and future networks, neural networks (NNs) and AI, quantum computing, quantum key distribution, and the intricate realm of lab-on-chip biosensors, we delve into the multifaceted impact of integrated optics across diverse technological frontiers.

4.1. Quantum key distribution (QKD)

Integrated optics is a transformative technology with wide-ranging applications. In data centres, it enables faster and more efficient information transfer by manipulating light on a Si chip and integrating various photonic elements onto a single chip for a compact and energy-efficient solution. In communication networks, integrated optics minimizes signal loss over long distances, while in quantum computing, it provides a platform for manipulating and controlling quantum bits [150]. The technology is instrumental in Quantum Key Distribution (QKD) [151–153], ensuring secure communication through the generation and recognition of quantum-entangled photons.

In today's digital society, the reliance on data security is paramount, encompassing not only secure communication but also the safeguarding of stored information in crucial domains such as e-banking, e-business, e-health, and e-government channels. However, with the advancing capabilities of quantum computers posing a potential threat to traditional security measures, the risk to individuals and the integrity of essential systems is becoming increasingly tangible. In the context of preserving sensitive health records, which often require lifelong security, QKD emerges as a pivotal technology uniquely equipped to address the challenge of long-term data security. Similarly, critical infrastructures like energy provision must confront the quantum threat, and quantum technologies are poised to play a pivotal role in fortifying

these vital systems [154].

Despite being commercially available for several years, ongoing efforts are dedicated to refining QKD systems, making them more compact, cost-effective, and capable of spanning longer distances. These developments are critical to fostering the widespread adoption of quantum technologies by both governmental bodies and industries. The current frontier lies in seamlessly integrating QKD systems into existing network infrastructures, presenting a complex challenge that requires collaborative efforts from multidisciplinary teams. These teams, comprised of telecommunication equipment manufacturers, critical infrastructure providers, network operators, QKD equipment providers, digital security professionals, and scientists, are collectively working towards a cohesive and resilient future for data security in the quantum era [154,155].

Si photonics stands out as an important platform for quantum photonic technologies, offering advantages such as miniaturization, cost-effective device manufacturing, and compatibility with CMOS microelectronics. Despite these benefits, the effective modulation of quantum states at high speeds within standard Si photonic platforms has been constrained. In addressing this limitation, the research presented in Ref. [156] successfully overcame the hurdle and showcased the achievement of high-speed, low-error QKD modulation using Si optical devices. This breakthrough involves a combination of slow TO DC biases and rapid carrier-depletion modulation with a BW of 10 GHz. The scalability of these integrated circuits, coupled with their incorporation into microelectronics, paves the way for the development of advanced quantum communication technologies. This innovation not only signifies a significant leap forward in the field but also holds promise for the broader adoption of quantum-secured communications.

To address challenges linked with the saturation and phase-dependent loss of high-speed carrier-depletion modulators (CDMs) in standard Si photonic manufacturing [156], an inventive approach was introduced. This approach involves the amalgamation of slower yet optimally performing thermo-optic phase modulators (TOPMs) with faster yet suboptimal CDMs for achieving QKD state preparation at GHz speeds. With the utilization of this hybrid technique, Fig. 5(a–c) illustrates three sample use cases of high-speed, low-error QKD: chip-to-chip coherent one-way (COW) QKD, polarization encoded BB84, and time-bin encoded BB84 state preparation. These experiments

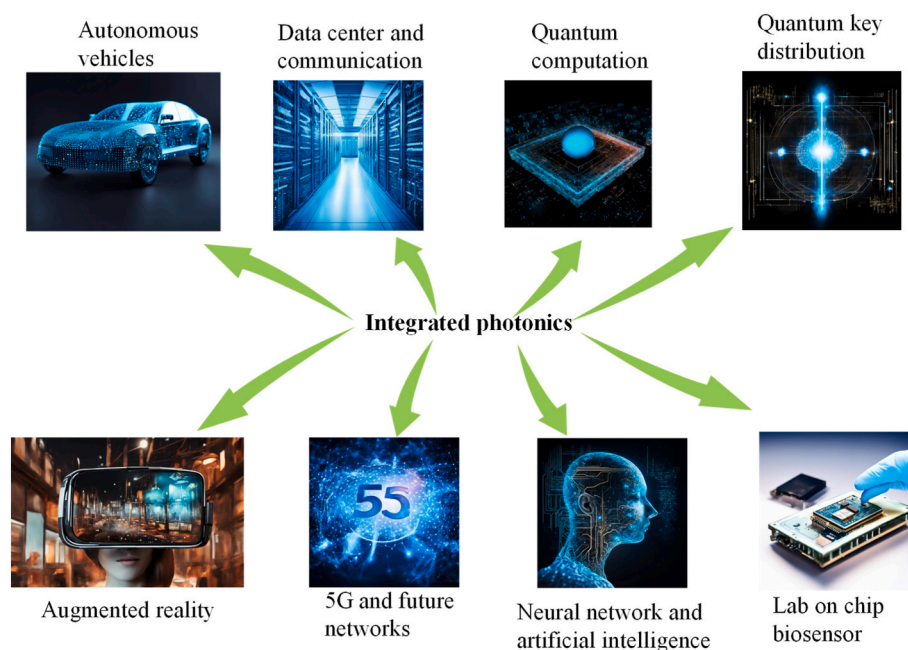


Fig. 4. Future of integrated optics in several domains of life.

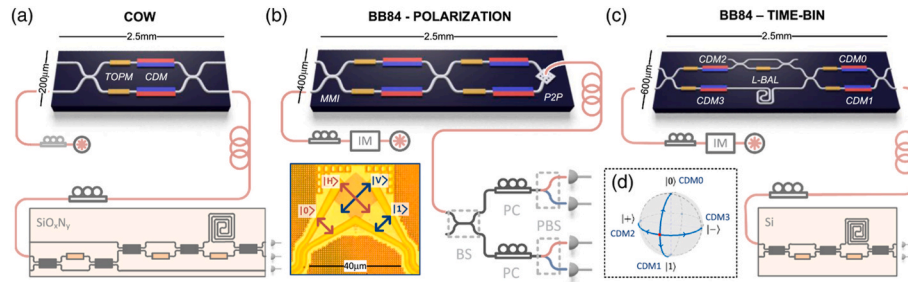


Fig. 5. Designed for QKD, Si optical devices are integrated into (a) a coherent-one-way system, employing a balanced MZI with MMI devices serving as beam splitters. Quantum information encoding through path or pulse modulation is achieved using TOPMs and CDMs [156]. In (b) Polarization encoded BB84, MZI paths are combined with a grating coupler to transform path-encoded information into polarization-encoded information, making it suitable for free-space (FS) communication [156]. (c) Time-bin encoded BB84 utilizes an unbalanced asymmetric MZI (AMZI) for temporal encoding, with coherent pulses separated into two intervals using on-chip delay. DC offsets set by TOPMs are illustrated on the Bloch sphere, and fast modulation by CDMs enables the encoding of each BB84 state. The overall design highlights a variety of strategies for implementing QKD in Si optical devices [156].

demonstrated impressive results, revealing projected asymptotic secret key rates reaching 916 kbps. Notably, they exhibited remarkably low quantum bit error rates (QBERs) of just 1.01 %, even when transmitting data over a 20 km fibre distance. These results emphasized the practicality of integrating high-speed QKD circuits into standard Si photonic manufacturing processes [156].

4.2. Quantum computation

Quantum computation signifies a groundbreaking shift in computing, leveraging the principles of quantum mechanics to execute computations in ways beyond the capabilities of classical computers. In contrast to classical bits confined to 0 or 1 states, quantum bits or qubits can exist in multiple states simultaneously through the phenomenon of superposition [157,158]. This property empowers quantum computers to concurrently process extensive amounts of information, potentially addressing intricate problems at an exponentially faster pace compared to classical counterparts. Another crucial aspect of quantum computation is the utilization of entanglement, wherein qubits become interconnected, enabling the instantaneous influence of one qubit's state on another, regardless of their physical separation. These distinctive features hold tremendous potential for tackling challenges in cryptography, optimization, and simulation that currently elude classical computing methodologies [159–161].

In the field of quantum communication, it is common practice to assemble systems using discrete optical elements. These elements, which are typically made of materials such as optical glasses (e.g., fused quartz and silica) and optical crystals (e.g., calcite, beta barium borate, and LN), are usually assembled separately and interconnected through free space or optical fibres [162]. While optimizing individual elements for specific requirements, for example ultra-low loss, high efficiency, fast speed, and high fidelity in quantum information applications, is convenient and adheres to the principles of conventional discrete optical designs, we recognize that challenges in interconnects and packaging persist. We believe this becomes especially pronounced when dealing with extensive networks connecting hundreds of thousands of clients. Ensuring high mechanical and thermal stabilities becomes crucial in order to address issues such as space and phase misalignment over time, resulting from environmental stresses and temperature variations. It is a significant challenge to achieve such stability in a complex discrete-optical system through global stabilization. Consequently, existing bulky systems comprising discrete optical modules may encounter difficulties in meeting the mounting need for higher-volume transmission capabilities. This highlights the significant advantages of chip-scale quantum communication systems.

The advent of quantum photonic chips has ushered in a new era of quantum technology. These chips offer distinct advantages over traditional methods, including miniaturization, scalability, and stability

[163,164]. Miniaturization is a notable strength of quantum photonic chips, as it a priori reduces the size of the chip, which is a crucial aspect in the development of quantum technology. Furthermore, the chips are crafted as a unified entity through lithography, which facilitates scalability. This contrasts with the sequential construction of individual elements, which is a common approach in the development of quantum technology. Not only does this streamline the manufacturing process, but it also allows for expansion in a seamless manner, a crucial aspect for large-scale quantum applications. The construction of circuits on a solid-state platform provides stability, which is a key asset. This characteristic minimizes deviations caused by vibrations or fluctuations in temperature, ensuring consistent and reliable performance. Such stability is paramount for getting the required level of integration and efficiency in quantum information processing and communication [165]. Likewise, quantum photonic chips exhibit robust potential for cost-effective manufacture. While the primary expense associated with fabricating the necessary photomasks is notable, the per-chip cost can be significantly lowered through mass manufacturing. This highlights the economic feasibility of leveraging quantum photonic chips, making them a compelling choice for advancing quantum technologies [163].

A fundamental constraint on the scalability of various photonic quantum information processing systems is the flux of quantum light within a PIC. The inherent limits of on-chip photon flux are determined by source brightness and WG loss. Despite substantial progress in achieving exceptionally low-loss chip-scale PICs and high-brightness single-photon sources (SPSs) individually, integrating these advancements has proven to be a formidable challenge.

Presenting a breakthrough, the successful incorporation of a quantum emitter SPS with a wafer-scale, exceptionally low-loss Si₃N₄ photonic circuit was demonstrated in Ref. [165]. In this accomplishment, triggered and pure single-photon emission into a Si₃N₄ photonic circuit was showcased, featuring a propagation loss of ~1 dB/m at a wavelength of approximately 0.93 μm. Additionally, these observations include resonance fluorescence in the strong drive regime, signalling promising strides toward coherent control of quantum emitters. These findings marked a significant step forward in the development of scaled chip-integrated photonic quantum information systems. This amalgamation is especially crucial for tasks such as storing, time-demultiplexing, or buffering deterministically produced single photons, thereby advancing the capabilities of quantum information processing on a chip.

Fig. 6 (a) depicts the graphic of a hybrid integration platform. The exceptionally-low-loss waveguides (ULLWGs) consisted of a high-aspect-ratio Si₃N₄ core buried beneath a 1 μm SiO₂ upper cladding layer. The thickness of the top cladding was selected to support propagation of a weakly-confined single TE-guided mode with low propagation losses in the 0.9 μm wavelength range. The on-chip SPS included a GaAs nano-WG with embedded InAs self-assembled quantum dots (QDs)

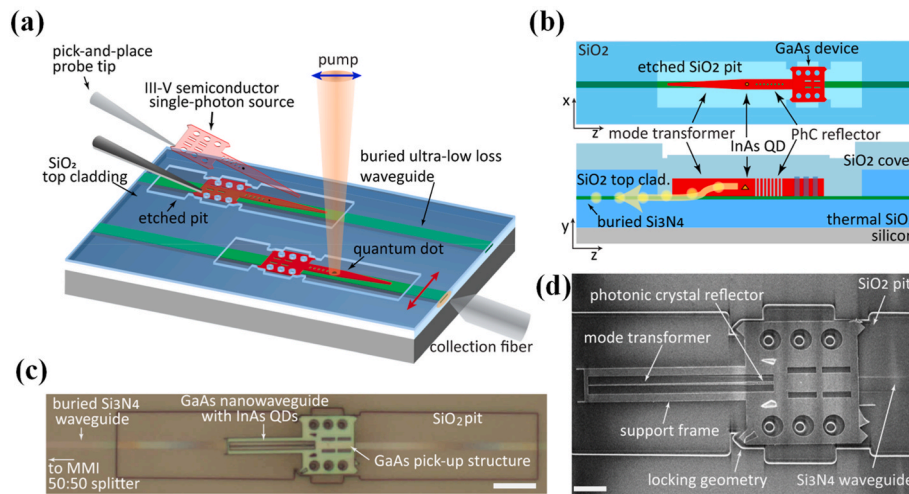


Fig. 6. (a) An illustration that attempts to show the pick-and-place hybrid integration process of a GaAs nanophotonic device, which houses InAs QDs, onto a ULLWG [165], (b) A schematic that attempts to showcase the top-view and cross-sectional [165] (c) Optical micrograph capturing a GaAs/InAs QD SPS assembled on a Si_3N_4 low loss WG, indicating a 50:50 MMI power splitter (not shown). The image was taken before the deposition of the top SiO_2 cladding [165], (d) Scanning electron micrograph offering a detailed view of the device before the deposition of the SiO_2 top cladding [165].

and, after that an adiabatic mode transformer. This configuration efficiently coupled QD emission into air-clad Si_3N_4 ridge WGs. Subsequently, an adiabatic taper was employed, succeeded by a 1-D PhC back-reflector designed for high reflectivity above $0.9 \mu\text{m}$, ensuring unidirectional emission into the Si_3N_4 WG. It is important to note that sections of the Si_3N_4 ULLWG, distant from the GaAs device, remained unaffected, as depicted in Fig. 6 (b) [165].

In Fig. 6 (c), an optical microscope image reveals the assembly of a GaAs nano-WG positioned above a buried Si_3N_4 ULLWG, leading into a 50:50 MMI splitter. The figure outlines the etched SiO_2 cladding corresponding to the placement pit for the GaAs device. The nano-WG geometry, housing the QD single-photon emitter, was enclosed by a frame designed for mechanical placement and structural support. It was connected to a pick-up pad, facilitating its transfer onto the Si_3N_4 chip. The SEM image in Fig. 6 (d) provides a detailed view, showcasing auxiliary locking features in the GaAs device geometry that complement the etched placement pockets. These features aid in alignment. From Fig. 6 (d), a misalignment of less than 340 nm between the GaAs device and the Si_3N_4 WG, as well as a tilt angle of less than 0.9° , can be deduced [165]. In summary, these findings suggested promising opportunities for employing quantum emitters as on-demand SPSs in exceptionally low-loss PICs with propagation losses of $\leq 1 \text{ dB/m}$. This development holds significant potential for the establishment of scaled photonic quantum information systems directly on-chip.

4.3. Neural network and artificial intelligence

NNs serve as a foundational element within the realm of AI, embodying a computational model encouraged by the interconnected neurons found in the human brain [166]. These networks are composed of layers of nodes, akin to artificial neurons, intricately arranged to process and analyze information [157]. Utilizing a training process, NNs acquire the ability to recognize patterns, make predictions, and execute tasks by fine-tuning the weights of connections between neurons based on input data. This adaptive learning capability positions NNs as a pivotal technology in the broader landscape of AI [167].

AI, in a broader sense, encapsulates the concept of machines or systems demonstrating intelligent behaviour, employing diverse methodologies. Within this scope, NNs play a vital role in machine learning, a subset of AI. The collaborative interplay between NNs and AI has resulted in significant advancements, powering applications such as image and speech recognition, as well as natural language processing.

This collaboration empowers machines to undertake intricate cognitive tasks and continually enhance their performance through experiential learning [168,169].

Photonic neural networks (PNNs) leverage the unique properties of photons to execute intelligent inference computing at an unprecedented speed while consuming ultra-low energy, making them ideal for ultra-high-throughput applications [170,171]. This innovation established a foundation for the next generation of intelligent computing, providing an efficient photonic hardware solution. This approach not only facilitates large-scale incorporation for on-chip, all-optical computing chips but also addresses the escalating demand for efficient computational power driven by the rapid advancements in PNNs [172]. Despite the astonishing progress in this field, challenges persist. The currently available nonlinear activation functions, based on optical nonlinearities, are often weak and impractical, hindering the full potential of PNNs. Furthermore, the absence of appropriate configurations for integrated photonic hardware and the lack of proper optical storage media presents additional hurdles. Confronting these challenges is essential for unlocking the full capabilities of PNNs and ensuring their seamless integration into the broader landscape of intelligent computing.

The conventional photonic computing framework, operating within the constraints of the von Neumann architecture, aims to create discrete optical devices with pre-optimized operation, with seamless integration into a PIC. This framework strives to develop a photonic computing chip that has regular physical structures and clear principles. However, it is constrained to a singular function and is burdened by substantial integrated crosstalk (CT). The inherent trade-off between achieving an ultrafast response time and maintaining ultralow energy usage presents formidable barriers to the large-scale amalgamation of standard photonic chips following the von Neumann architecture, given that this architecture has been designed for processing tasks with specific functions. In contrast, the emerging photonic computing framework, particularly the NN model, unfolds within the non-von Neumann framework, which is structured yet limited in nature. This innovative approach liberates the system from the constraints of traditional architectures, providing flexibility and adaptability beyond singular functions. The NN model's departure from the von Neumann architecture signifies a paradigm shift, opening new avenues for more versatile and efficient photonic computing systems. This innovative paradigm embraces characteristics such as processing multiple functions in memory and executing intelligent inference tasks. Unlike its von Neumann counterpart, the NN-based photonic computing framework

demonstrates versatility and adaptability, paving the way for enhanced computing capabilities beyond singular, predefined functions. The shift towards a non-von Neumann architecture represents a paradigmatic leap, showcasing the potential for more dynamic, multifunctional, and intelligent photonic computing systems.

Over the past few decades, various approaches to PNNs have been explored [173,174]. Initially, before the advent of integrated WGs made possible by advancements in CMOS fabrication, optical neural networks (ONNs) predominantly took the form of FS diffractions. The implementation of FS ONNs relied on spatial light modulators (SLMs) as a crucial element, serving as the programmable input light controller for the NN. This configuration facilitated analogue computation with negligible CT noise, marking the inception of optical computing, featuring innovations like optical matrix multipliers and optical diffractive-NNs [175]. However, the practical deployment of FS ONNs faced challenges due to the reliance on SLMs and the need for free space. In contrast, integrated photonic neural networks (IPNNs) emerged as a more compact alternative for achieving high integration [172]. Nonetheless, IPNNs grapple with their own set of challenges, including weak optical nonlinearity, a lack of suitable configurations for photonic hardware, and the absence of proper optical storage mediums.

Integrated photonic deep neural networks (IPDNNs) have become widely utilized in diverse fields, ranging from computer vision to medical diagnosis [176]. Typically, these networks leverage clock-based processors [177], where computation speed is primarily constrained by clock frequency and memory access time. Despite recent strides in photonic computation in the optical domain, the scalability of optical deep networks is hindered by the absence of scalable on-chip optical nonlinearity and the inherent losses in optical devices. These limitations were addressed in Ref. [170] by introducing an integrated end-to-end

IPDNN. This groundbreaking system achieved image classification at sub-nanosecond speeds by directly manipulating optical waves within the on-chip pixel array as they traverse layers of neurons. Inside each neuron, precise optical linear computations took place, complemented by the optoelectronic implementation of non-linear activation functions. This distinctive approach resulted in remarkable classification times under 570 ps, aligning with the rapid pace of a single clock cycle on cutting-edge digital platforms. The scalability of the IPDNN was a key strength, bolstered by a uniformly distributed supply of light, ensuring a consistent optical output range for each neuron. This thoughtful design not only facilitated the extension of IPDNNs to larger scales but also enhanced their overall efficiency. Validation of the system's efficacy was demonstrated through two-class and four-class classification of handwritten letters, realizing impressive accuracies surpassing 93.8 % and 89.8 %, respectively. A critical advantage lies in the system's direct, clock-less processing of optical data, eliminating the need for analogue-to-digital conversion and bypassing the necessity for a large memory module. This innovation resulted in faster and more energy-efficient NNs, poised to drive forward the next generation of deep learning systems [170]. The 30 signals collected by the input pixels undergo a sophisticated process, wherein they were organized into 4 sets of overlapping 12-pixel sub-images. These sets were then individually directed to specific neurons in the first layer, labelled I_1 to I_4 . This intricate routing was facilitated by a photonic distribution network, incorporating cutting-edge nanophotonic WGs, Y-junction splitters, and WG crossings, as illustrated in Fig. 7 (b). For the training of the IPDNN chip and calibration of image formation, a secondary on-chip component came into play—a 5×6 grating coupler array, mirroring the configuration of the input array. This component served a crucial role in the process. Each pixel in this array captured the optical power it

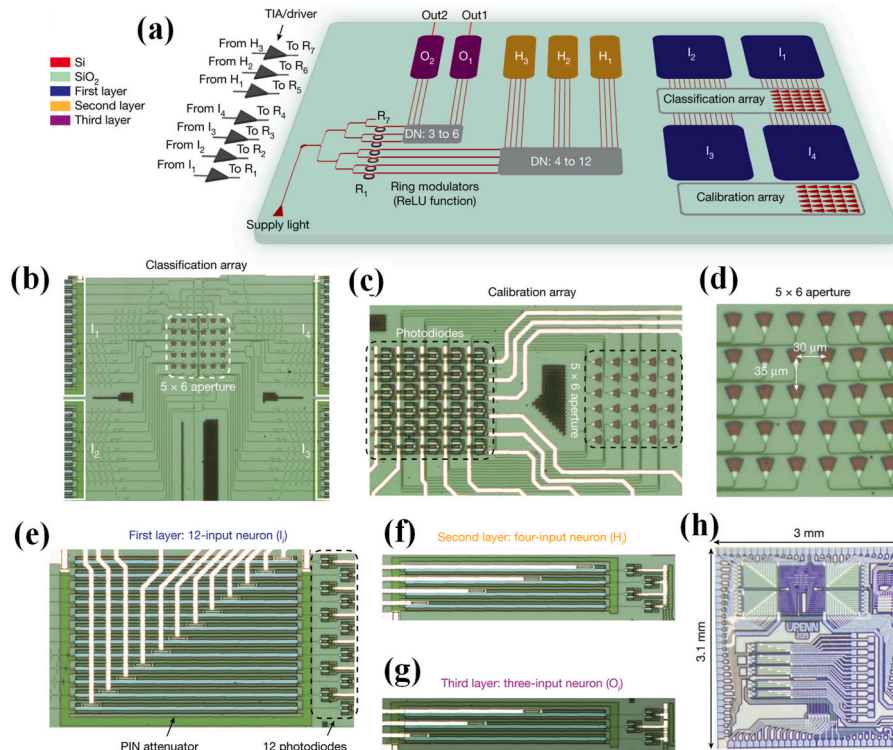


Fig. 7. Primary block diagram of the PDNN chip (a). The input pixel array (b) and the calibration array (c) are made up of two 5×6 arrays of grating couplers [170], (d) greater detail about the 5×6 grating coupler array, with a particular focus on the corresponding element pitch. The input pixel array, which plays a crucial role in classification, produces four sets of 12 optical signals that are directed to the neurons in the first layer. The distributed light is evenly spread across neurons in the second and third layers, traversing seven MRMs to implement the ReLU non-linear activation function. Seven off-chip TIAs drive the on-chip modulators. The system produces two outputs, which allows for up to four class classifications [170]. The microphotographs in (e–g) provide an insight into the individual neurons within the first, second, and third layers, showcasing the PIN attenuators and parallel PDs positioned after the attenuators [170]. In (h), a microphotograph of the photonic chip realized in the AMF 180-nm SOI process is presented. DN represents the distribution network [170].

received, and this data was observed using a dedicated photodetector, as depicted in Fig. 7 (c). A closer look at the setup is presented in Fig. 7 (d), demonstrating a microphotograph of the 5×6 -pixel array, featuring an aperture size of approximately $140 \mu\text{m}$ by $150 \mu\text{m}$. This glimpse into the intricate design highlights the precision and advanced technology at play in signal processing and analysis within this system [170].

Once the input pixels had been organised into overlapping sub-images for convolution, the light underwent processing through three layers. The initial layer (see Fig. 7 (e), input layer), which consisted of 12-input neurons (I_1) and established complete connectivity with the second layer (hidden layer), which was composed of four 12-input neurons (H_1), was followed by the third layer (output layer), which comprised two three-input neurons (O_1). The outputs of I_1 to I_4 and H_1 to H_3 were linked through TIAs/drivers (R_1 to R_4 and R_5 to R_7 , respectively) to the MRMs, supporting up to four-class classification. The third layer (output layer, see Fig. 7(g)) comprised two three-input neurons (O_1). The outputs of I_1 to I_4 and H_1 to H_3 were linked to MRMs R_1 to R_4 and R_5 to R_7 through TIAs/drivers, respectively, which allowed for the classification of up to four classes. To facilitate the supply of light to individual neurons across all layers, a laser was integrated into the IPDNN chip. Fig. 7 (h) presents a microphotograph of an IPDNN chip, realized in the AMF 180-nm SOI method [135], which offers insight into the chip's design [170].

4.4. Data centre and communication

At the core of modern-day information technology, the data centre serves as a vital hub for the storage, processing, and management of extensive digital data. Within this technological nexus, the communication infrastructure stands as the essential lifeblood, guaranteeing seamless connectivity and effective data exchange [178]. Robust communication networks are indispensable for data centres, linking servers, storage systems, and other pivotal elements, enabling them to collaborate harmoniously in handling diverse tasks [179]. Swift data transmission is facilitated by high-speed and reliable communication protocols between servers and storage devices, ensuring the prompt execution of computing operations and the accessibility of information [180]. The intricate interplay of networking technologies, including Ethernet and fibre optics, ensures the fluid flow of data within the data centre's ecosystem [181]. The ongoing evolution of communication technologies plays a pivotal role in enhancing the efficiency and responsiveness of data centres, ultimately shaping the modern form of computing and information management.

The cumulative storage capacities of global data centres now exceed a staggering 13 Exabytes, a quantity equating to approximately 1 billion Gigabytes per Exabyte for context. This colossal figure is poised for continuous expansion, fuelled by emerging opportunities in the realms of the Internet of Things (IoT), Big Data, and AI. Confronting this escalating demand, the combination of optic fibres and integrated optics emerges as a transformative solution [182]. At the forefront of this technological paradigm shift are PICs, microscopic chips harnessing the power of photons—light particles—as opposed to traditional electricity. This innovative approach empowers PICs to sense, process, and transmit data at the astounding speed of light [183]. From facilitating communication between individual chips on a circuit board (PCB) to orchestrating long-haul transmissions for interconnecting data centres, the versatility of PICs knows no bounds. The combination of optic fibre communication and PICs represents a pivotal leap forward in enhancing the performance of data centres. By leveraging these advanced technologies, data centres can realize significant improvements in efficiency, speed, and overall functionality, thereby meeting the escalating demands of our data-driven world [184]. The synergy of optic fibres and PICs stands as a beacon of progress, illuminating a path towards a more robust and responsive digital infrastructure [185].

InP PICs provide a robust avenue for consolidating progressive routing and signal-processing functions onto a single chip.

Demonstrating impressive throughputs at line rates of 320 Gbps within multistage networks, these circuits achieved circuit connectivity scaled to 16 ports. Beyond, scalability in connectivity is attainable through on-chip wavelength routing. To unlock essential cost-performance benefits, achieving higher circuit densities, efficiencies, and improved polarization handling requires additional combination and the introduction of new elements. Meeting the demands for lowered energy utilization, lower latency, and decreased costs amid rising data rates necessitates ongoing advancements in sub-system integration. While the analog nature of the photonic elements poses a challenge, abstraction is crucial to enable practical network-level implementations. Key functionalities such as multicast capability, packet timescale reconfigurability, power levelling independent of the signal path, distributed signal quality surveillance, and data rate transparency can be executed at the chip level. This achievement paves the way for an abstracted digital control plane [186].

The potential impact of attaining this level of signal processing extends beyond immediate concerns in data communications, offering prospects for applications ranging from sensor readout technologies to optical beam-forming technologies. As the pursuit of enhanced performance and efficiency continues, the combination of InP PICs stands as a transformative force shaping the future landscape of advanced signal processing in various technological domains. Separating the photonic physical layer from the control plane was essential to effectively showcase packet routing in integrated optical switch matrices [186]. The most efficient solution was to use fast current driver arrays in conjunction with a round-robin scheduler. Fig. 8(a) illustrates a 4×4 broadcast-and-select switching matrix designed to validate packet time-scale connectivity. The data were fed into each of the 4 inputs, and time traces were documented for all-optical outputs. A round-robin scheduler was employed to execute diverse path combinations sequentially within a $3.2 \mu\text{s}$ time slot. As depicted in Fig. 8 (b), the output time trace from the 1st output showcases independent level control for each semiconductor optical amplifier (SOA) gate, enabling on-chip optical power levelling. Notably, the entirety of the circuitry required only a single clock input signal to coordinate with the external network, with multiple gate states defined concurrently, thereby facilitating the comprehensive utilization of the fabric. This integrated approach marked a significant leap forward in achieving efficient packet routing within optical switch matrices, laying the foundation for improved connectivity and control in OCSs [186].

Efficiently configured high-radix 4×4 SOA-based switch elements were pivotal in multistage networks, providing substantial connectivity with a limited number of stages. Balancing optical wiring complexity, signal integrity, and control intricacy, three switch stages represented an optimal compromise. This concept was demonstrated with a monolithic 16×16 connectivity switch, realized with active-passive integration, as shown in the composite microscope photograph in Fig. 8 (c). The input and output WGs, visible on a $250 \mu\text{m}$ grid at the left and right edges of the circuit, were integral components of the design. The circuit featured an input stage on the left with 2×2 switch arrays and an output stage on the right. These were seamlessly integrated through two planes of 4×4 switch arrays. The 4×4 switches, equipped with loss-compensated transmission and select capability, were demonstrated to support multicast functionalities at both the switch element and circuit levels. The circuit, one of the most sophisticated of its kind, contained 192 active SOA gates and 188 passive splitters and combiners, occupying a compact area of $4.0 \times 13.2 \text{ mm}$. Its design allowed connections from any port to any other port, requiring only three SOA gates on any given path. An additional switching plane at the central stage enhances flexibility, providing two distinct paths for each input-output combination and facilitating an adaptable nonblocking design. This achievement marked a significant advancement in the complexity and functionality of high-radix SOA-based switch elements within optical communication networks [186].

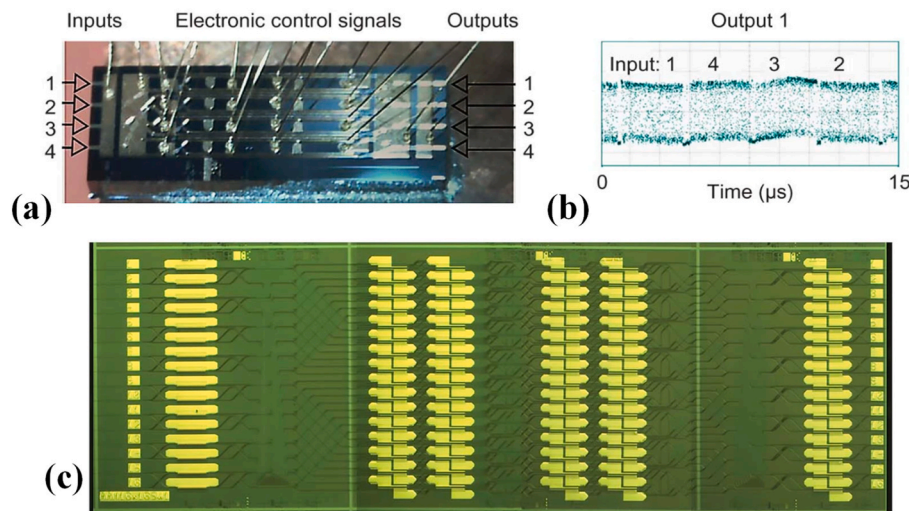


Fig. 8. (a) 4×4 broadcast-and-select switch in operation, utilizing a round-robin scheduler, illustrating [186], (b) nanosecond switch reconfiguration for optical packet routing [186], (c) 16×16 output photonic switch matrix [186].

4.5. 5G and future networks

Integrated optics is crucial for advancing 5G and future communication networks, supporting higher data rates, lower latency, and the implementation of advanced technologies like wavelength division multiplexing (WDM) and coherent communication [34,187–192]. Additionally, it plays a pivotal role in neuromorphic computing and ONNs, leveraging light's unique properties for efficient information processing in parallel, contributing to the evolution of AI. The increasing digitalization, mobile Internet connectivity, and AI applications have made data immensely valuable. The exponential increase in data creation and consumption necessitates higher data transfer rates in both wired and wireless domains. The existing 5G infrastructure, however, falls short of accommodating the swiftly growing demand for data and extensive connectivity due to its BW limitations.

To overcome these challenges, there is a pressing need to transition to terabit-per-second (Tbps) connectivity, which is a primary objective for the next generation of communication networks, particularly 6G and beyond. Terahertz (THz) frequency bands are anticipated to play a critical role in the development of future high-speed wireless networks, including 6G and 7G, enabling ultra-high BW broadcasts [193]. The extensive spectral BW of THz waves is vital for establishing Tbps communication links without necessitating additional measures to boost spectral efficiency. In addition to their wide BW, THz waves possess unique attributes, such as a short wavelength that facilitates highly directional communication links, thereby enhancing security against eavesdropping. THz waves also exhibit reduced susceptibility to free space diffraction and inter-antenna interference, which allows for the integration of transceivers and antennas into much smaller devices [194]. Moreover, THz radiation may be formed into so-called pencil beams, i.e., beams of ultranarrow beamwidth and high directionality that can be emitted from, e.g., leaky or MIMO-array antennas [195], which can further be utilized in 3D position estimation. These properties make THz waves indispensable for achieving the ambitious goals of 6G and beyond, which aim to seamlessly integrate the physical, biological, and digital realms [196]. THz waves are crucial for providing pervasive connectivity, low latency, better energy efficiency, advanced sensing capabilities, and precise 3D positioning, marking a significant advancement in communication technologies.

An innovative leap in THz technology was highlighted with the unveiling of a pioneering category of on-chip, phototunable devices [194]. This progress encompasses a comprehensive single-channel 160 Gbit/s communication link seamlessly integrated with a demultiplexer

(DEMUX) based on Silicon Valley PhC. The pivotal achievement was centred around the optically controllable demultiplexing capabilities, precisely designed for two distinct carriers' modulated signals, all accomplished without any undesired CT. This capability was enabled through the utilization of topologically protected states, achieved with a critically coupled high-quality cavity. As a validation of the concept, the system effectively demultiplexed extraordinary spectral efficiency 40 Gbit/s signals and showcased real-time broadcasting of uncompressed high-definition (HD) video at an impressive 1.5 Gbit/s, leveraging the state-of-the-art topological photonic chip.

This innovative technology, known as phototunable Si topological photonics, was set to revolutionize the field of CMOS-compatible terahertz (THz) technologies. Its importance lies in its ability to propel the advancement of future 6G and 7G communication eras, enabling real-time terabit-per-second (Tbps) wireless connectivity. This advancement is crucial for a wide range of applications, including network sensing, holographic communication, and the cognitive Internet of Everything, driving transformative progress towards unprecedented levels of wireless communication and connectivity. Two distinct devices were pivotal in this study: Device-1, which combined straight Valley Photonic crystal and bend Valley Photonic crystal WG chips, and Device-2, an on-chip topological demultiplexer (DEMUX). Optical images of the fabricated straight Valley Photonic crystal and bend Valley Photonic crystal chips are shown in Fig. 9 (a), with the domain walls of the Valley Photonic crystal chips marked by blue-shaded rectangles. A single-channel broadcast link achieved data rates of 160 Gbps and 125 Gbps using the straight Valley Photonic crystal and bend Valley Photonic crystal chips, respectively. The inset of Fig. 9 (a) displays the I-Q constellation diagram for the 160 Gbps data rate using 32 quadrature amplitude modulation (QAM-32) with a 32 GBaud symbol rate for both the straight Valley Photonic crystal chip and the reference. The well-isolated points in the diagram indicated minimal distortion in the transmitted signal, highlighting the robust data signal passage through the Valley Photonic crystal chip. This demonstrated the effectiveness of the straight Valley Photonic crystal and bend Valley Photonic crystal chips in maintaining signal integrity at high data rates [194].

Fig. 9 (b) illustrates an on-chip topological demultiplexer (DEMUX) featuring a Valley Photonic crystal WG precisely coupled to a high-quality topological cavity. Optimizing the distance between the WG and cavity is essential for achieving maximum CE. This setup enabled the frequency demultiplexing of a single channel into two well-separated channels with minimal crosstalk. Fig. 9 (c) shows the arrangement used for the empirical measurement of terahertz

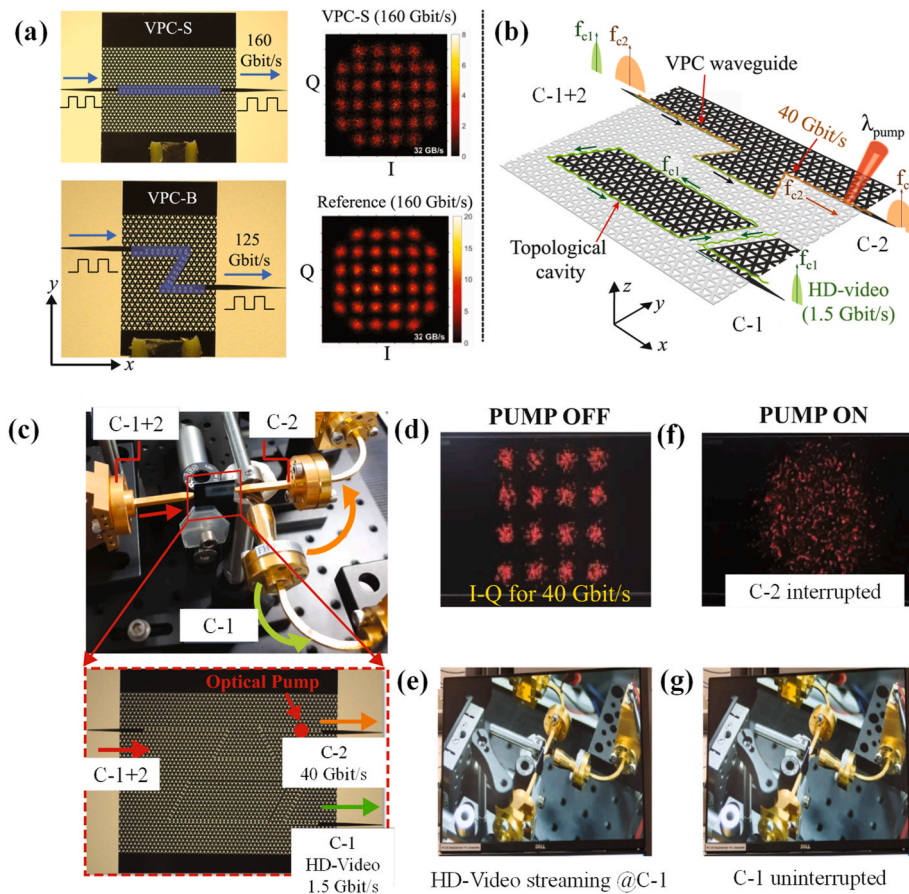


Fig. 9. (a) Optical image of assembled straight Valley Photonic crystal and bend Valley Photonic crystal Si-Valley Photonic crystal chips [194], (b) illustration depicts the graphic of an on-chip topological DEMUX. Two data signals, modulated with distinct schemes at carrier frequencies f_{c1} and f_{c2} , are introduced into the Valley Photonic crystal WG. Subsequently, the DEMUX efficiently separates these signals into two distinct channels: C-1 and C-2 [194], (c) empirical arrangement for THz broadcast, (d, f) I-Q intensity constellation diagram captured at C-2 of the Si-Valley Photonic crystal chip under both PUMP-OFF and PUMP-ON conditions. In the absence of photoexcitation (PUMP-OFF), a distinct constellation diagram is evident, representing a 40 Gbit/s data rate using QAM-16 with a 10 GBaud symbol rate. Conversely, with photoexcitation (PUMP-ON), the THz waves experience attenuation, leading to a fading effect in the constellation diagram [194], (e, g) Capture from live HD video broadcasting through C-1 in both PUMP-OFF and PUMP-ON scenarios. Notably, the continuous real-time broadcasting of HD video remains unaffected even after diminishing the THz signal at C-2 during photoexcitation [194].

demultiplexing. The input taper port of the Si-Valley Photonic crystal chip was placed between the dual-channel transmitters. The two outputs of the Si-Valley Photonic crystal chip, acting as a frequency demultiplexer, are directed to two distinct receivers. The first receiver (C-1) detected the amplitude-modulated signal carrying the HD video signal, using direct recognition for real-time signal transmission. The second receiver (C-2) employed a sub-harmonic mixer for phase/amplitude signal recognition, which was crucial for high spectral efficiency modulations such as quadrature amplitude modulation (QAM). The inset in Fig. 9 (c) provides a detailed view of the Si-Valley Photonic crystal chip, highlighting the labelled input and output ports. The carrier frequency of 331.6 GHz was modulated employing ON-OFF keying, while the carrier frequency of 344 GHz utilized QAM-16 modulation at a rate of 10 GBaud. Positioning C-2's carrier frequency far from cavity resonance allowed access to a substantial BW of 14 GHz [194].

In Fig. 9 (d), the I-Q constellation diagram corresponds to QAM-16 with a symbol rate of 10 GBaud and an occupied spectrum of 14 GHz. The topological cavity resonance facilitated the demultiplexing of the input data into two channels with an exceptional isolation ratio. To demonstrate this capability, C-2 underwent photoexcitation, resulting in signal attenuation and a fading constellation diagram (refer to Fig. 9 (f)), while HD-video broadcasting in C-1 remained unaffected (see Fig. 9 (e) and (g)). This observation affirmed the incredible isolation between demultiplexed channels, ensuring that undesired broadcasted power

from C-2 to C-1 does not disrupt the video broadcasting process [194].

4.6. Lab-on-chip biosensors

In the field of lab-on-chip biosensors, IO offers a compact and highly sensitive platform for the rapid recognition of biological molecules [197]. By integrating various photonic elements onto a single chip, these biosensors can perform biochemical assays in a miniaturized and portable format, enabling real-time, label-free sensing [35]. This miniaturization enhances sensitivity and specificity, making it valuable for applications in medical diagnostics, environmental surveillance, and point-of-care testing [198]. The ability to perform complex assays on a miniaturized scale holds great promise for point-of-care testing, enabling quicker and more cost-effective diagnostics, and ultimately contributing to improved patient care and public health [199]. As technology continues to advance, lab-on-a-chip biosensors are poised to play a pivotal role in revolutionizing the landscape of biomedical and analytical sciences [198,200,201].

Over the last thirty years, there has been a noticeable upswing in the advancement of biosensor devices, finding applications in diverse fields such as environmental surveillance, biotechnology, medical diagnostics, drug screening, food safety, and security. While optical biosensor technology has reached a high level of maturity with various commercially available products, its widespread application in real-world scenarios

has been hindered by challenges related to stability, sensitivity, and size. The emergence of integrated photonic biosensors, grounded in Si technology, signifies a transformative breakthrough that holds the promise to surmount these limitations. These innovative biosensors stand as a compelling solution, providing advanced diagnostic tools marked by heightened sensitivity, specificity, and overall reliability. This heralds a new era in biosensing technology, paving the way for enhanced performance and broader utilization of critical applications [202]. This breakthrough holds the promise of enhancing the efficacy of both in-vivo and in-vitro diagnostics, addressing existing limitations, and opening the door for more robust applications in various practical scenarios.

In the field of detecting evanescent waves, a receptor layer is strategically fixed onto the core surface of the WG. The deliberate exposure of the modified surface to analyte molecules that complement it triggers a biochemical interaction. This interaction results in a specific modification in the optical characteristics of the biological layer. The nuanced alteration is detected through the evanescent field of the guided light, and its magnitude can be accurately linked to both the analyte concentration and the interaction's affinity constant. This association delivers a quantitative signal, providing a valuable understanding of the molecular interaction's nature and strength. The array of optical sensing platforms spans a wide spectrum, encompassing sensors based on optical fibres [203], planar WG structures [204], micro-resonators [144,205], resonant WG diffractive structures [206,207], light-addressable potentiometric devices (LAPS) [208,209], porous Si [210], and more. Each platform contributes its unique strengths, fostering a diverse landscape of tools and techniques that cater to a myriad of applications in the burgeoning field of optical biosensing.

The merits of optical sensing are notably elevated when employed within the framework of IO technology which facilitates the incorporation of both passive and active optical elements—such as fibres, emitters, detectors, WGs, and related devices—onto a single substrate. This incorporation allows for the agile development of compact and miniaturized sensing devices, offering the added capability to produce multiple sensing devices on a single chip. The integrated approach offers a multitude of advantages, including miniaturization, robustness, reliability, mass production capabilities with a subsequent reduction in production costs, low energy consumption, and straightforward alignment of individual optical elements.

Among the selection of technologies available for photonic biosensor fabrication, Si technology stands out as one of the most practical and promising tools. Extensive work on Si optical devices within the telecommunications sector has yielded successful applications for sensor development [18]. Within the field of "siliconized" photonics, device fabrication utilizes Si or Si-related materials employing microelectronics technologies. This approach aims to consolidate all sensing modules onto a single chip. The process of "siliconizing" photonics involves investigating key building blocks, including light generation, and coupling, selective guidance and transport within WGs, light encoding, detection, device packaging, and, crucially, the "smart" electronic control of these photonic functions. Notably, the biofunctionalization of the sensing element constitutes a distinct area of research, underscored as a separate block in the diagram of Fig. 10.

Integrated biosensors play a vital role in environmental monitoring by offering high sensitivity, real-time analysis, and multiplexing capabilities for detecting diverse contaminants. Leveraging innovative technologies like plasmonic or TriPlex™ waveguides and versatile structure of integrated multichannel fluorescent biosensors, enable the simultaneous detection of multiple analytes with exceptional accuracy [211,212]. For instance, the TriPlex™ waveguide chip provides a means for monitoring 32 individually addressable sensor patches, allowing parallel analysis of multiple analytes with detection limits as low as 0.22 µg/L for MC-LR, 0.2 µg/L for atrazine, 1.18 µg/L for 2,4-D, and 0.06 µg/L for BPA [211]. Furthermore, the reusable and scalable designs enhance cost-efficiency and throughput, making fluorescence biosensors

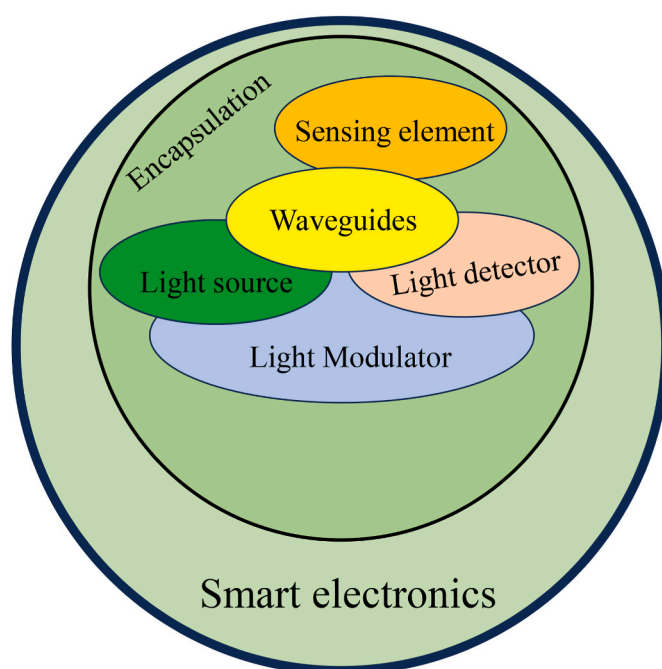


Fig. 10. Key focus areas in advancing Si technology-based photonic biosensors.

indispensable for rapid, precise, and sustainable monitoring of environmental pollutants in real-world scenarios.

Advancements in consistent and early diagnosis methods for cellular diseases are imperative. A promising avenue for achieving this is the combination of a biological selectivity probe, such as an aptamer, as a key recognition layer for the recognition of crucial biomolecules. Notably, lung cancer remains a leading cause of cancer-related fatalities, underscoring the urgency of innovative diagnostic approaches. In Ref. [213], the development of an electrical sensing platform designed for the specific and label-free recognition of a human lung carcinoma cell line (A549) was presented. Fig. 11 (a) shows the manufactured microchip. The platform used amine-terminated aptamer-modified Au electrodes, forming a robust foundation for precise molecular recognition. The microdevice incorporated a coplanar electrode configuration and a straightforward microfluidic channel on a glass substrate. Fabrication employed standard photolithography and cast molding techniques, with a proposed self-assembly procedure onto the Au surface. Observations through optical microscopy and measurements using electrical impedance spectroscopy validated the microchip's capability to selectively and effectively identify A549 cells. Notably, the dominant capacitance element in impedance changes was calculated at an optimal frequency, offering a robust basis for evaluating biosensor sensitivity. This pioneering approach yielded a biosensor that is simple, cost-effective, biocompatible, and highly selective. With its potential to specifically detect early-stage lung cancer, this biosensor stood for a significant stride towards advancing diagnostic methodologies for cellular diseases.

The illustrated empirical arrangement in Fig. 11 (b) clarifies the trial procedure. Electrochemical Impedance Spectroscopy (EIS) measurements were conducted across a broad frequency spectrum using a Wayne Kerr 6420 impedance analyzer. BNC cables connected the aptasensor electrodes to the device. Solution injection into the chip channel was carried out with a syringe pump. Cellular samples were observed using a fluorescence microscope featuring a charge-coupled device camera, seamlessly linked to a computer running Olympus DP Controller image software. The recorded observations were then effortlessly transferred to the computer through a digital interface. For the electrical data, a smooth transition occurred to the computer via the same digital interface. The intricate impedance parameters of the microchip were

subsequently subjected to thorough analysis employing the sophisticated LabVIEW software. It is noteworthy to highlight that all impedance measurements were conducted under controlled conditions at room temperature within a 10 mM PBS buffer solution. This standardized and controlled environment ensured the reliability and accuracy of the impedance data, providing a robust foundation for the interpretation and conclusions that were drawn from the experimental results.

Diverse A549 cell samples were employed in the experiments to estimate the sensor's capabilities. The sensor's capacitance was estimated based on the analysis of the imaginary part of the impedance extracted from EIS data. In the graphical representation in Fig. 11 (c), the capacitance profiles show fluctuations across various cell concentrations, ranging from 1×10^5 to 5×10^5 cells/mL. A reduction in capacitance magnitude was observed with increasing cell concentration, particularly within the low-frequency range. The variation in capacitance was quantified as the deviation from the baseline capacitance value, with the baseline corresponding to the chip without cells but featuring aptamer-modified Au electrodes. An exceedingly linear correlation between capacitance changes and cell concentration was observed at a frequency of 5 kHz. The precision of this correlation was exemplified in the linear regression equation presented in Fig. 11 (d), showcasing a correlation coefficient (R^2) surpassing an impressive 99 %. To determine the limit of detection (LOD), formula $3\sigma/\text{slope}$ was utilized, where σ represents the standard deviation, and the slope was derived from the linear response. In this context, the sensor demonstrated a commendable LOD at approximately 1.5×10^4 cells/mL, underscoring its sensitivity and reliability for detecting subtle deviations in cell concentrations [213].

Additionally, PIC based photonic sensors have also emerged as a transformative technology in environmental and agricultural monitoring, offering unparalleled precision and efficiency [214,215]. By leveraging light-based detection mechanisms, these sensors can measure a wide range of parameters, such as air quality, water purity, soil composition, and crop health, with high sensitivity and accuracy [216]. In agriculture, photonic sensors enable real-time monitoring of plant conditions through optical spectroscopy, identifying stress factors like drought, nutrient deficiencies, or pest infestations early [217]. Similarly, in environmental applications, they can detect trace amounts of pollutants or greenhouse gases [218,219], contributing to improved ecosystem management and pollution control. Their non-invasive nature, coupled with the ability to integrate into IoT systems, makes

photonic sensors an essential tool for advancing sustainable practices in both sectors [220].

4.7. Autonomous vehicles

Integrated optics is proving to be a game-changer within the field of Autonomous vehicles (AVs), offering transformative solutions to critical challenges in their development [221]. At the core of this technological convergence is the role of photonics in refining key elements such as Lidar systems, where the miniaturization and integration of photonics elements enhance the precision and efficiency of distance measurement. Beyond this, integrated optics contribute to the creation of high-speed optical broadcast networks within AVs, fostering real-time data exchange between onboard systems [222].

Autonomous driving stands poised to transform road traffic, mitigating existing challenges such as accidents and congestion. Over the years, a concerted effort from car manufacturers, researchers, and administrations has propelled significant advancements in autonomous driving technology. Despite these strides, formidable obstacles remain, spanning not only intricate automotive technologies but also encompassing the orbits of human behaviour, ethics, traffic management strategies, policies, and liability [221]. The complexity of implementing an autonomous driving environment has prompted a cautious approach among carmakers, deferring the commercial launch of fully driverless vehicles in the short term. One of the paramount technical challenges lies in achieving unequivocal obstacle detection at high speeds and over extended distances. This hurdle underscores the need for continuous innovation in sensor technologies and computer vision systems.

Within the field of traffic management, consensus points toward a vision where vehicles collaborate harmoniously. Various strategies, such as General Vehicle-to-Vehicle (V2V) cooperation and platooning, are actively under consideration, each with its nuanced variants [223]. Simulation plays a fundamental role in the design and validation of these strategies, offering a controlled environment for testing diverse scenarios. However, the journey towards autonomous driving is not solely a technical one. Legal and ethical dimensions have already surfaced, adding layers of complexity to the landscape of highly automated driving. These concerns range from the necessity of specialized driving licenses to intricate topics like determining liability in the event of an accident and addressing privacy issues. As the technology becomes more feasible, these legal and ethical considerations could potentially impede

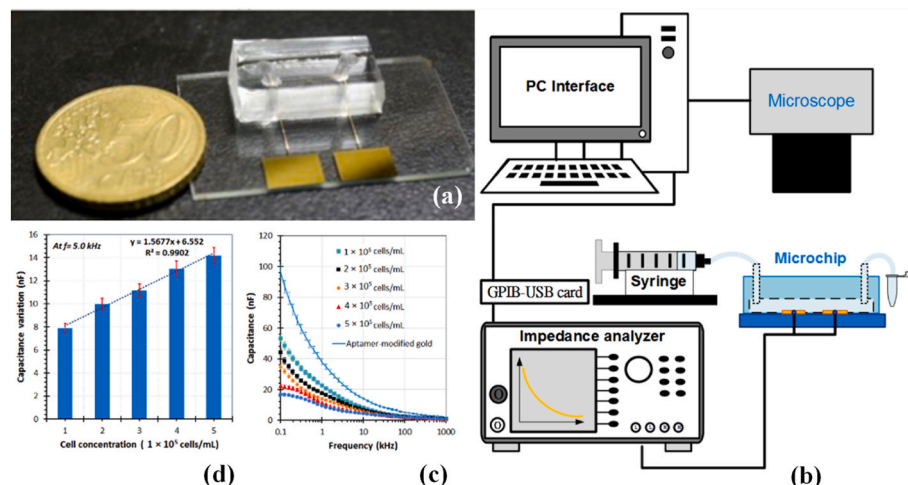


Fig. 11. (a) Image of a manufactured microchip created through the conventional photolithography procedure [213], (b) illustration of the empirical setup featuring an inverted microscope and an impedance analyzer managed through the LabVIEW software interface [213], (c) analysis of the capacitance element in the impedance response across various A549 cell samples confined on the aptamer-based sensing electrodes of the microchip. This assessment was conducted through the impedance analyzer at different cell conc. [213], (d) examination of the capacitance alteration in the chip with varying cell conc. concerning the aptamer-modified electrodes, measured at a frequency of 5 kHz [213].

the widespread adoption of AVs. Balancing innovation with a comprehensive understanding of societal, ethical, and legal implications remains imperative for steering the future of autonomous driving on a safe and responsible course.

PICs mark a significant advancement in addressing the hurdles associated with cost-effective large-scale production and attaining reliability suitable for automotive applications. The key lies in incorporating optical elements such as lasers and detectors directly onto Si chips [224]. This advancement not only allows to decrease size and cost but also to harbor the possibility of scalable mass production. Silicon photonics introduces an entirely new level of system integration, paving the way for the creation of solid-state, frequency-modulated, continuous-wave (FMCW) LiDAR-on-Chip.

FMCW LiDAR technology, characterized by its unique operational principle and lower laser power demands in comparison to time-of-flight (ToF) LiDAR, is widely recognized as a crucial sensor for facilitating autonomous mobility [225]. Its compatibility with integrated photonic modules makes it especially well-suited for progress in the field. The co-integration of PICs and EICs was facilitated by cutting-edge system-in-package tools, such as flip chip and 3D chip stacking. This harmonious blend results in a synergistic combination of efficient and potent photonic and electronic performance [226].

An additional advantage lies in the elimination of fibre, a component traditionally associated with high costs and vulnerability to the challenging environmental conditions encountered by vehicles daily on the road. This removal enhances the overall robustness and durability of LiDAR systems integrated with PICs and contributes to the realization of a new era in automotive sensing technology [227]. The convergence of PICs and LiDAR technology not only addresses critical challenges in manufacturing and reliability but also propels the automotive industry toward a future where autonomous mobility is seamlessly integrated and technologically advanced [226].

Scantinel Photonics is at the forefront of innovation, actively advancing the development of a cutting-edge FMCW optoelectronics core module (OCM) built upon CMOS-compatible photonic integration platforms [228]. At its heart, the module boasts solid-state scanning PICs, as illustrated in Fig. 12 (a). Notably, Scantinel has achieved a significant milestone by achieving parallelization of multiple FMCW channels, a testament to the module's exceptional capabilities. The LiDAR system adeptly handles adverse weather conditions, including fog, rain, and snow, with exceptional effectiveness (Fig. 12 (b)). Moreover, its coherent range remained impervious to interference from other LiDAR systems or external light sources, such as sunlight. Notably, this advanced system provided instantaneous acquisition of direct velocity information for each measured pixel. The flawless integration of these sophisticated features elevates safety awareness to an unprecedented level, setting a new standard in LiDAR technology.

Through precise engineering and rigorous testing, Scantinel has successfully demonstrated the module's capability to deliver consistent and dependable scanning performance. This achievement was a

testament to the dedication and expertise of the Scantinel Photonics team [228]. Adding to its prowess, the module was complemented by an advanced optical system, further enhancing its scanning capabilities. The synergy between the solid-state scanning PICs and the sophisticated optical system culminates in outstanding scanning results. This powerful combination ensured that the module excels in various applications, meeting the demands of diverse ecosystems. What sets this module apart is its compact modular design, facilitating seamless integration into a myriad of application ecosystems. This adaptability was a key feature, allowing the module to be effortlessly incorporated into a wide range of scenarios. As a result, Scantinel's FMCW OCM not only pushes the boundaries of scanning technology but also provides a versatile and user-friendly solution for applications across industries.

As the demand for compact, efficient, and cost-effective solutions in mobility applications continues to grow, the synergy between PICs and microelectronics becomes a catalyst for innovation. The assimilation of photonic and electronic functionalities on a single chip not only streamlines manufacturing processes but also positions Si photonics as a key enabler for advancing LiDAR technologies. This confluence of technological advancements propels the mobility sector into a future where sophisticated sensing capabilities are seamlessly integrated, fostering safer, more efficient, and intelligent transportation solutions.

4.8. Augmented reality (AR)

Integrated optics is also being employed in the evolving field of AR, enhancing the immersive experiences and functionality of AR devices. By integrating several photonic parts, for instance lasers, detectors, and WGs, onto a single chip, integrated optics aids in the expansion of compact and efficient optical systems crucial for AR applications [229]. These devices empower the manipulation of light to project digital information seamlessly into the user's physical environment [230]. In AR glasses or headsets, integrated optics facilitates the creation of vibrant and realistic holographic displays, providing users with interactive and dynamic overlays of the real world. This technology enhances fields like gaming, education, healthcare, and navigation, offering users a more natural and engaging AR experience [231]. As integrated optics continues to progress, it is poised to play an integral role in the ongoing advancement of AR, contributing to the development of sleeker, more powerful, and versatile AR devices that seamlessly balance the virtual and physical worlds [232].

The burgeoning field of Si photonics in FS applications, which spans LiDARs, FS optical communications, and quantum photonics, underscores the need for advanced emission-shaping capabilities that surpass the conventional limits of grating couplers [233]. In these scenarios, Si photonic chips assume a crucial role in precisely directing emissions into free space for the detection or manipulation of external objects. Effectively generating light emission from a Si photonic chip into free space requires specific spatial modes, serving diverse purposes such as focusing, collimation, orbital angular momentum (OAM), and

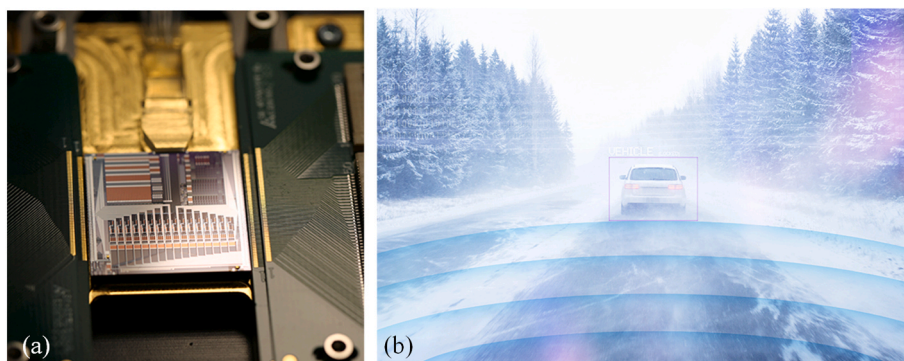


Fig. 12. (a) Solid-state PIC scanner [228], (b) Immune to interference [228].

holographic projection [234,235]. Spatial modes define the distribution of the electric field's amplitude and phase in emitted light and are tailored using various on-chip and external devices. Grating couplers, for instance, employ periodic structures to couple light from waveguides into free space, with their geometry dictating beam direction, divergence, and polarization [236]. Meta-optics and diffractive optical elements (DOEs) further enable precise control over light by introducing subwavelength structures to create advanced spatial profiles like OAM beams or holographic wavefronts [237,238]. Similarly, microlenses, either integrated or separately fabricated, are used to focus or collimate light, often in conjunction with gratings for optimized coupling [239]. Photonic crystal structures, with their wavelength- and mode-selective properties, can generate directional emissions or exotic spatial modes, while holographic beam shapers encode complex wavefronts for imaging or projection [240,241]. Finally, tapered waveguides are employed to transition light smoothly from confined photonic modes to broader free-space modes, ensuring efficient coupling and minimal distortion [242,243]. These devices collectively enable diverse and precise control over light emission for a wide range of photonic applications.

To address this demand, a cutting-edge platform is introduced, designed not only to facilitate the versatile shaping of FS emission but also to retain the essential attributes of CMOS compatibility and monolithic integration inherent in Si photonics. This innovative approach not only expands the scope of Si photonics applications but also enhances the efficiency and adaptability of FS optical technologies, establishing a new paradigm for the seamless integration of photonics in various real-world scenarios. Li et al. proposed an integrated OAM beam emitter featuring radially polarized radiation, implemented on a shallow-ridge silicon microring with azimuthally distributed gratings etched atop the ring waveguide [244]. Two critical structural parameters—the number of grating elements and the dimensions of each grating—were optimized to maximize OAM order purity (C_p) and total radiation efficiency (η_{all}). Through meticulous design, simulations demonstrated a C_p of approximately 98 % and an η_{all} of about 49 %. Additionally, the calculated energy ratio between the far-field radially and azimuthally polarized components exceeded 11 dB, indicating the dominance of the radially polarized component in the emitted electromagnetic field [244].

This innovative platform seamlessly integrated metasurfaces (MSs) onto PICs, featuring amorphous Si nanopillars evanescently coupled to Si WGs. MSs represent engineered two-dimensional structures designed

for the precise manipulation of EM waves [245]. In contrast to conventional optical modules like lenses and mirrors, MSs employed subwavelength-sized structures to control the phase, amplitude, and polarization of light across a planar surface. This innovative approach facilitated the development of compact, lightweight devices with versatile optical capabilities [246]. Typically, MSs consist of an arrangement of subwavelength resonators, such as nanoantennas or dielectric scatterers, organized in specific geometric patterns. The careful design of these patterns empowers MSs to exert unprecedented control over the propagation of light, achieving outcomes beyond the reach of traditional optics [247].

The empirical validation underscored the remarkable capabilities of diffraction-limited beam focusing, achieving an impressive Strehl ratio of 0.82. The dynamic control exhibited by the focused spot was noteworthy, enabling seamless switching between two positions through precise manipulation of the excitation direction. An integrated meta-hologram (MH) served as a compelling demonstration of the exceptional versatility inherent in the MS platform within Si photonics [233]. The MH design, featuring dimensions of 40 μm by 40 μm , projected a distinct image of the letter “ ψ ” with pinpoint accuracy at a height of 20 μm above the MS. The application of the Gerchberg–Saxton algorithm facilitated the determination of the requisite phase profile, vividly illustrated in Fig. 13 (a). To further validate the design, a comprehensive full-wave 3D FDTD simulation was executed, revealing in Fig. 13 (b) a simulated holographic image of the letter “ ψ ” strictly constructed at the specified distance above the MH. This empirical evidence solidified the efficacy and precision of the proposed methodology.

Optical microscope and SEM photos of the assembled MH are displayed in Fig. 13 (c) and 13 (d), affirming the successful realization of this device. The measured intensity distribution at 26 μm above the MS, illustrated in Fig. 13 (e), closely aligned with the simulation results, providing empirical validation for the MH's performance [233]. This experiment highlighted the platform's versatility in generating radiation with intricate wavefronts, surpassing the capabilities of standard grating couplers. The on-chip holographic projection exhibited in this experiment holds promising applications, particularly in near-eye displays for virtual/AR and single-pixel imaging. This innovative assimilation of MHs on Si optical devices not only advanced the capabilities of FS emission but also opened new avenues for transformative applications in emerging technologies [233].

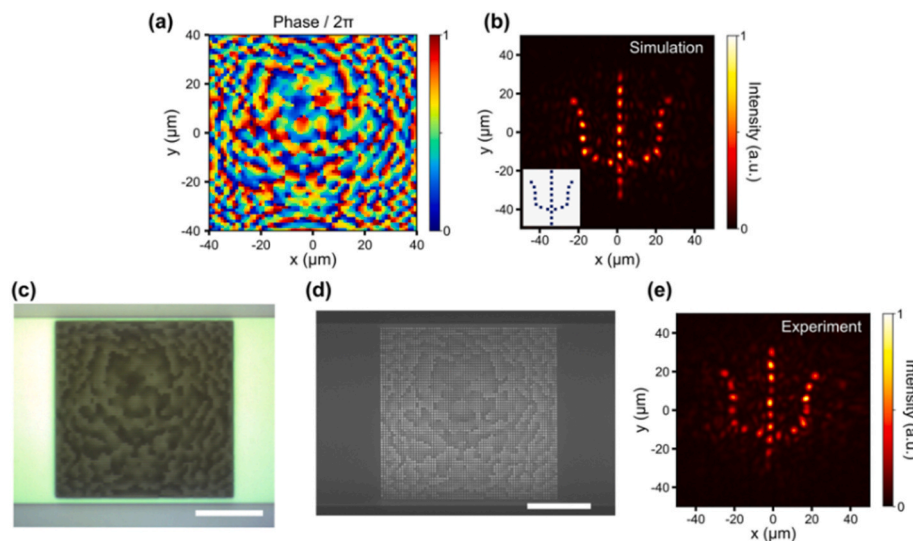


Fig. 13. Simulation and practical validation of an integrated MH. (a) Generation of the phase profile through the Gerchberg–Saxton algorithm, enabling the projection of the letter “ ψ ” at 20 μm above the MS [233]; (b) FDTD-simulated intensity mapping at the specified distance above the MS, with an inset displaying the target image [233]; (c) Optical microscope image capturing the details of the manufactured MS [233]; (d) SEM image providing a closer look at the intricacies of the fabricated MS [233]; (e) Empirical measurement of the intensity mapping at 26 μm above the MS, validating the real-world performance of the MH [233].

5. Navigating challenges: exploring viable solutions for a resilient future

Acknowledging the considerable advancements in photonic integration over the past 10 years, it's undeniable that this technology still lags three decades behind electronics [248]. Numerous areas within this field necessitate further development and maturity. Evolution, however, must be approached from a business viewpoint. Optical constituent and system markets naturally gravitate towards cost reduction, enhanced functionalities and performances. Yet, establishing the right ecosystem of customers and providers requires time, paving the way for profitable ventures for both small start-ups and large corporations [249].

Presently, despite the availability of capabilities at reasonable prices from various providers, numerous challenges persist for end-users embarking on photonic integration projects. The primary challenge lies in comprehending photonic integration technology itself, encompassing its current capabilities and limitations. This endeavour is particularly demanding due to the constant and dynamic development of the current technological possibilities and the creation of new photonic cutting-edge concepts, which may be a game-changer soon. To fully leverage the gains offered by PICs, optical engineers must adapt their mindset, akin to the evolution witnessed in electronic incorporation. The complication of potential optical systems, combined with cost efficiencies at higher volumes and probable for drastic miniaturization, opens doors to applications and implementations previously inconceivable.

Yet, newcomers may find it difficult to comprehend these ideas, and a comprehensive grasp is necessary before embarking on PIC design. Critical factors, including intellectual property (IP) limitations, variations among suppliers in the supply chain, and the transition from prototype to pilot production or mass production, can significantly impact project outcomes. Consulting with an experienced partner in the field can help mitigate these initial challenges. Additionally, establishing effective interaction between the photonic integration sector and prospective end-users can pose difficulties. While the telecom sector has traditionally dominated the PIC market, the same level of understanding may not be present in other photonic integration markets, such as sensing, quantum, or medical optics, where customers may possess diverse technical backgrounds and languages. Successful interaction is vital for comprehending customer needs along the value chain, emphasizing the value of a partner capable of bridging the gap between both sides.

While the appeal of cost-effective integrated optics technology is evident due to its economic compensation, it is essential to find such solutions with a discerning perspective. The lower price often entails compromises in terms of precision, reliability, and overall performance. Integrated optics devices with budget-friendly pricing may utilize less advanced materials and manufacturing processes, potentially making them more susceptible to environmental factors, limiting their operational lifespans, and compromising functionality [27]. Applications that demand cutting-edge photonics solutions, such as telecommunications, sensing, or quantum computing, may necessitate the superior performance and stability provided by higher-end alternatives [250]. Lately, Butt et al., in collaboration with consortium partners from Poland, confronted challenges associated with the development of an inexpensive optical platform based on silica-titania materials [47,144, 251–254]. Consequently, users and industries delving into integrated optics should be aware that opting for low-cost options may not consistently meet the desired level of efficiency and reliability required for advanced and demanding applications. Balancing affordability and quality become a necessity when steering the landscape of integrated optics technology.

Regarding the design component, elements like the IP of design building blocks and fabrication processes carry substantial weight for customers beforehand. There is a persistent demand for more automated and standardized software design tools, and customers can profit from

unified frameworks that cover design elements ranging from mode solving to mode propagation, circuit simulation, and layout. This is particularly vital due to the restricted compatibility among software tools and the elevated costs linked to licensing. Initiatives like the PDAFlow Foundation [255], focused on establishing software guidelines for photonic design automation, along with the availability of rental software licenses and design houses, contribute to addressing these challenges.

In the fabrication domain, there remains a perceived shortage of up-to-date and comprehensive information regarding foundry capabilities, schedules, and pricing. Considerable room for improvement exists in the adoption and optimization of process design kits (PDK) for generic production processes, encompassing the establishment of consistent building block performances and tolerances [256]. Additionally, delays are prevalent, and assurances on manufacturing in multi-project wafer run still fall short. While there is progress in photonic foundries offering manufacturing reports and enhanced tracking, these practices are not yet widely adopted.

Packaging remains a significant obstacle to photonic integration in integrated circuits (ICs) [257]. The challenge persists in reducing costs during both the prototyping and production stages. Although notable progress has been made, efforts are ongoing to enhance this aspect. Characterization packages and demo boards have emerged, enabling customers to authenticate proof-of-concept PICs and showcase early prototypes. However, there is a need for additional work on standardizing packages and designing templates that align with them. From a customer standpoint, the most critical information gap involves clearer narratives of capabilities, packaging options, and tolerances [258].

Testing poses a significant, often underestimated challenge in photonic integration projects, spanning bare dies, wafers, or packaged devices [259]. While akin to quality control at a production level, photonic integration necessitates early functional testing, including bare die characterization, intricate optical and electrical measurements, and statistical sampling. Achieving a higher degree of automation is essential to reduce testing costs and time, given the substantial investment in measurement equipment. Outsourcing testing can expedite the process and alleviate initial financial burdens. Challenges also arise in environmental qualification and certification, particularly in industries such as telecom, space, and military, where standards are less advanced than in electronic integration. In navigating the business landscape of photonic integration, common challenges persist, including cost escalation, production transfer, performance guarantees, and lead times. However, once these challenges are identified and adequate support is secured, companies can develop and produce disruptive PICs that have the potential to penetrate existing markets and create new opportunities.

6. Key insights and final thoughts

Integrated optics, by combining various photonic components on a single chip, is a key player in modern technology with momentous implications for daily life. It enables the manipulation of light for diverse applications, remarkably in telecommunications where it supports high-speed data transmission over optical fibres, crucial for an interconnected world. Beyond communications, integrated optics has transformed sensing technologies, benefiting medical diagnostics, environmental surveillance, and industrial processes. In healthcare, it has led to compact and effective diagnostic tools, while in computing, it shows promise for faster and more energy-efficient data processing. The ongoing evolution of integrated optics advocates its impact on various fields, including AR, AVs, and quantum computing, highlighting its vital role in influencing the future of technology and improving the effectiveness of our everyday devices.

The evolution of photonic platforms for the development of PICs has been marked by a continuous quest for improved performance, miniaturization, and scalability. Early efforts focused on traditional semiconductor-based platforms, utilizing materials like Si, which

provided a solid foundation but faced challenges such as high-power expenditure and limitations in BW. As the field progressed, novel materials and structures, including Si photonics, emerged to address these issues. Si photonics offered improved integration capabilities and compatibility with existing electronic technologies. Beyond Si, researchers explored alternative materials like III-V semiconductors, polymers, and even two-dimensional materials like graphene. These diverse platforms enabled the assimilation of numerous photonic components, such as lasers, modulators, and detectors, on a single chip, facilitating the creation of more compact and efficient PICs. The continuous development of PIC platforms is still facing new challenges and, shortly, may experience breakthroughs that will shake up the current PIC landscape. One of the most promising paths to unveil a new realm of possibilities in integrated optics is the integration of widely understood metaphotonics, which may unlock new, previously unavailable, potential in terms of optical signal processing and storing [260–264] as well as on-chip optical computing [265–267], outweighing possible economic risks connected with the adaptation of existing or creation of parallel foundry frameworks to fit subtle requirements of metaphotonics. This perpetual evolution of photonic platforms continues to push the boundaries of what is possible in terms of data transfer rates, energy efficiency, and the combination of photonics into diverse applications, thereby shaping the future of optical communication technologies.

Advancing PIC technology necessitates a comprehensive strategy that encompasses research, innovation, and collaboration across various fronts. Initial emphasis should be placed on consistent investment in foundational research to explore innovative materials and manufacturing methods, thereby improving the performance and scalability of optical devices. Simultaneously, encouraging interdisciplinary cooperation among physicists, materials scientists, and electrical engineers is crucial for achieving breakthroughs in understanding and overcoming current limitations. Moreover, there is a need to actively promote standardization initiatives, establishing common frameworks for the design, manufacturing, and testing of PICs to facilitate widespread industry adoption. Expanding education and training programs is vital to cultivating a skilled workforce proficient in propelling innovation in the field. Furthermore, fostering partnerships between industry and academia, as well as encouraging international collaborations, can expedite the translation of research findings into practical applications, fostering the combination of photonic technologies across diverse sectors. In essence, a comprehensive and well-coordinated effort spanning research, education, standardization, and collaboration is indispensable for advancing PIC technology to unprecedented levels.

Abbreviations: Silicon-on-insulator = SOI; Subwavelength grating = SWG; Photonic integrated circuit = PIC; Electronic integrated circuit = EIC; Neural network = NN; Artificial intelligence = AI; Augmented reality = AR; Autonomous vehicle = AV; Process design kits = PDKs; Optical neural network = ONN; Spatial light modulator = SLM; Free-space = FS; Single photon source = SPS; Quantum dot = QD; Cross talk = CT; Ultra-low-loss waveguide = ULLWG; Waveguide = WG; Ring resonator = RR; Optical communication system = OCS; Quantum Key Distribution = QKD; Frequency-modulated, continuous-wave = FMCW; Optoelectronics core module = OCM; Coupling efficiency = CE; Metahologram = MH; Thermo-optic = TO; Demultiplexer = DEMUX; Bandwidth = BW; Lithium niobate on insulator = LNOI; Indium Phosphide = InP; Electro-optic = EO; Intellectual property = IP; Multi-chip module = MCM; Semiconductor optical amplifier = SOA; Terahertz = THz.

CRedit authorship contribution statement

M.A. Butt: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **B. Janaszek:** Writing – review & editing, Writing – original draft, Software, Resources, Project administration, Data curation. **R. Piramidowicz:** Supervision, Software,

Resources.

Conflict of interest

The authors declare no conflict of interest.

Data availability statement

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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